



Neurosplash

2023 한국뇌신경과학회 고등인지분과 하계 워크숍

일시

2023. 8. 20(일) ~ 21(월)

장소

청풍리조트 컨벤션홀



한국뇌신경과학회
The Korean Society for
Brain and Neural Sciences

ibS 기초과학연구원
Institute for Basic Science



Brain & Cognitive Sciences
서울대학교 뇌인지과학과

Neurosplash

2023 한국뇌신경과학회 고등인지분과 하계 워크숍

일시

2023. 8. 20(일) ~ 21(월)

장소

청풍리조트 컨벤션홀



한국뇌신경과학회
The Korean Society for
Brain and Neural Sciences

ibS 기초과학연구원
Institute for Basic Science



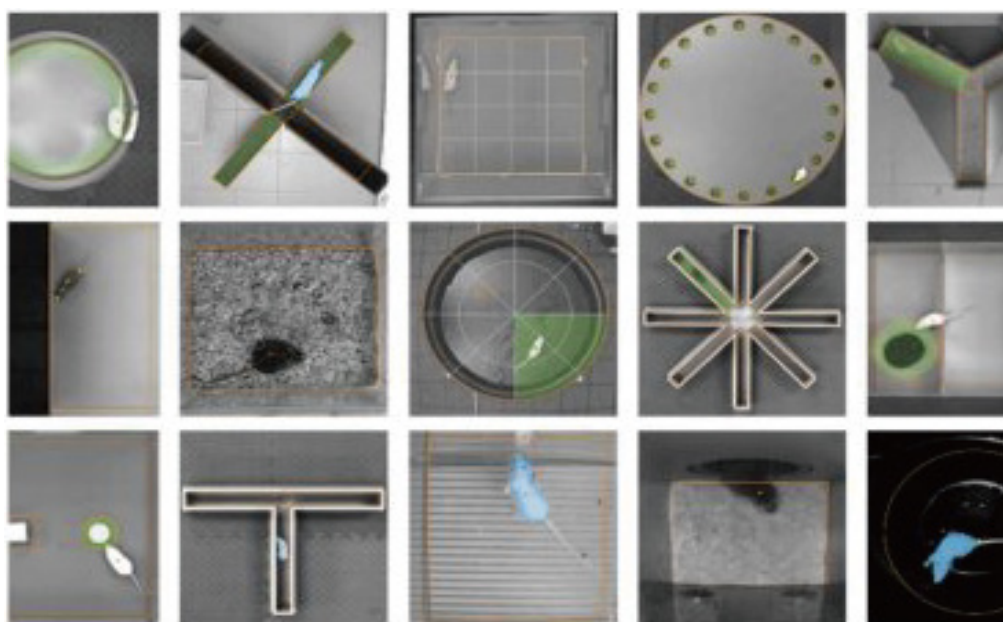
Brain & Cognitive Sciences
서울대학교 뇌인지과학과

Powerful Video Tracking Software

Use Literally ANY-maze



- ☑ **Zones:** Draw shapes to divide your apparatus into areas, then combine areas to create any number of zones
- ☑ **Sequences:** Define sequences to detect movements between areas or zones
- ☑ **Immobility and freezing:** Automatically detect immobility and freezing, perfect for fear conditioning



La Pias Ear Tags

5 Different Colors · Mouse & Rat · Durable · Easy Placement



- ☑ Ear tags are numbered from 1-100
- ☑ (available up to 999)
- ☑ Available as Stainless Steel
- ☑ (non-colored)
- ☑ or in 5 colors (1.Red 2.Green 3.Blue
- ☑ 4. Light Blue 5.Black)
- ☑ Once applied, they cannot be easily removed
- ☑ Laser-etched number provides durability

Animal Markers



- ☑ Nontoxic and devoid of any
- ☑ hazardous materials*
- ☑ A single marker can be used
- ☑ to mark 300-500 rats
- ☑ (depending upon animal size
- ☑ and/or applied area)
- ☑ Once applied, colors remain
- ☑ for 6-12 weeks
- ☑ Broad and fine tip markers available

Digital PCR

No.1 21년 10대 나노기술 선정!

No.2 Digital PCR 의료기기 NEP 인정!

No.3 Digital PCR 의료기기 식약처 인허가 획득!

No.4 COVID-19 진단키트 美 FDA 긴급사용 승인!



제품 동영상

Droplet generator consumable devices
Model : TGR-100



Droplet analyzer
Model : TAS-100



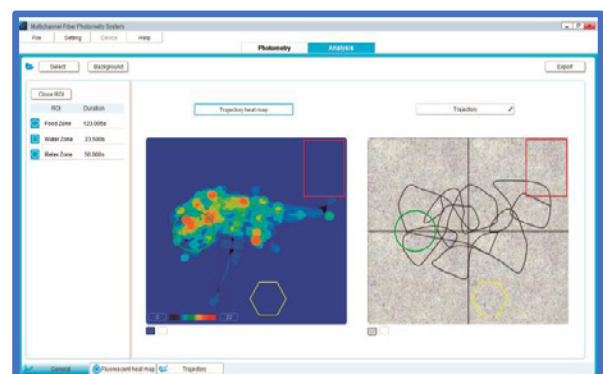
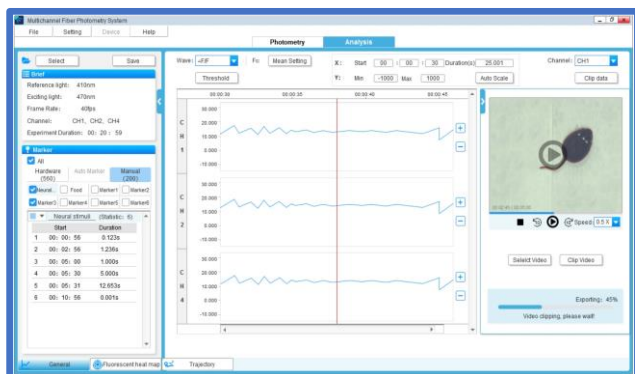
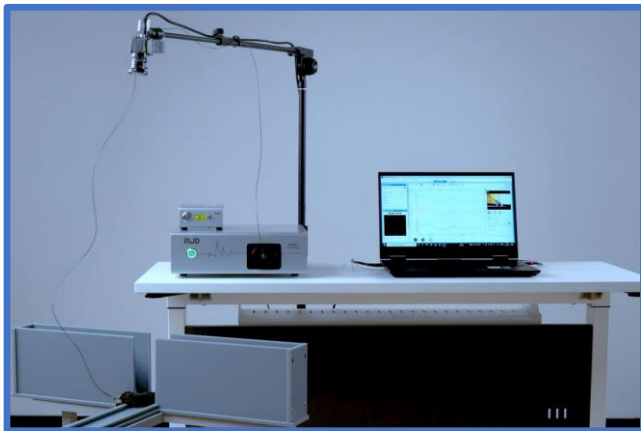
Fiber Photometry system

RWD

RWD R821

자유롭게 움직이는 동물의 뉴런 활동 측정

하나의 software 로 Brain 형광 데이터 및
동물 행동 비디오를 동시에 수집 분석 가능





2023 한국뇌신경과학회 고등인지분과 하계 워크숍

일시

2023. 8. 20(일) ~ 21(월)

장소

청풍리조트 컨벤션홀

1박 2일로 개최되는 본 하계 워크숍에서는 한국뇌신경과학회의 정식 분과로 출범한 고등인지분과 회원들이 모여 연구실의 최신 연구 주제와 방향을 소개하고 논의하는 기회를 마련함으로써, 고등인지 신경과학 분야의 최신 트렌드를 이해하고 서로 유용한 정보를 공유하고자 합니다.

일자	시간	프로그램
08월 20일	14:00-14:10	개회사
	14:10-14:40	포스터 홍보 타임
	14:40-17:40	포스터 세션
	17:40-18:00	기념사진 촬영 및 체크인
	18:00-20:00	저녁 만찬
08월 21일	10:00-10:30	분과 운영회의
	10:30-11:00	시상식 및 폐회사

P-01

Distractors modulate long-term memory storage

Hansol Lee¹, Jayoon Choi¹, Min-Suk Kang^{1,2*}

¹Department of Psychology, Sungkyunkwan Univertisy, ²Center for Neuroscience and Imaging Research, Institute for Basic Science

Keywords: working memory, long-term memory, distractor

The working memory means the ability to maintain and manipulate information in mind and the resistance against distracting stimuli is regarded as its hallmark feature. However, mechanism that maintains target representation despite the presence of distractor remains underspecified. One line of studies suggest that our brain relies more on long-term memory with increasing cognitive load and this change occurs slowly. We advance this research by evaluating whether we also relies on long-term memory only when stimuli are presented in a distracting context. Specifically, participants completed a working memory task with a unique set of stimuli without distraction (baseline block). A distraction block follows such that a half of stimuli are presented with distractors (expectation trials) and the other half without distractors (distractor trials). After completing the working memory task, an incidental memory tasl follows without notice. We found that we rely more on long-term memory without distractors if stimuli are presented in a distraction context compared to baseline. In addition, if we compare the two trial types (expectation and distractor trials), the long-term memory usage continues to increase in the distractor trials. These results suggest that our brain copes against distraction adaptively by utilizng long-term memory storage.

This research is supported by NRF-2022R1A2C2007363.

P-02

Inactivation of the intermediate hippocampus impairs value-based navigation in a virtual environment more severely compared to the dorsal hippocampus

Hyeri Hwang, Seung-Woo Jin, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Hippocampus, Goal-directed navigation, Virtual reality

During goal-directed navigation, the hippocampus needs to process spatial and value information of the goal location; yet, its underlying neural mechanisms are poorly understood. Based on the anatomical projections with valence-coding subcortical areas, we hypothesized that the dorsal hippocampus (dHP) and the intermediate hippocampus (iHP) should play differential roles in processing place and its associated value representation. To dissociate these subregions, body-fixed rats ($n=8$) were trained for a virtual reality place preference task with bilaterally-implanted cannulae in both dHP and iHP. The rat started a trial at the center of a circular platform, with his head orienting to one of the six start directions. The trial ended when he arrived in one of the two unmarked reward zones, each associated with a different amount of honey water at a six-to-one ratio. The well-trained rats preferred the high-reward zone, exhibiting a stereotypic left turn at the beginning of navigation. Inactivation of either subregions with muscimol increased travel distance ($F(2,14)=21.629$, $p<0.001$; one-way repeated-measures ANOVA), but only iHP-inactivated rats showed impaired place preference ($F(2,14)=10.562$, $p<0.01$). We measured the accuracy of wayfinding behavior by the departing direction at the start location and the arrival direction (i.e., the angle at which the rat first touched the unmarked peripheral boundary of the arena). A direct comparison between the dHP and iHP inactivation conditions showed that the iHP inactivation induced more severe impairment in efficient value-dependent navigation than dHP inactivation (p 's <0.05 for the mean vector length; paired t -test; $p<0.001$ and $p<0.01$ for the direction distribution for departure and arrival, respectively; Kuiper's test). Our findings suggest that the dHP is more important for accurate spatial navigation to the target location per se, whereas the iHP is more critical for finding the goal location with higher value.

the National Research Foundation of Korea Grants 2018R1A4A1025616, 2019R1A2C2088799, 2021R1A4A2001803, 2022M3E5E8017723, the BK21 program, and the Global Ph.D. Fellowship program

P-03

Human Memory Retrieval in Spatial Navigation

Min-Jae Jo¹, Ji-Yeon Jeong², Min-Suk Kang^{2*}

¹Department of German Language and Literature, Sungkyunkwan University,

²Department of Psychology, Sungkyunkwan University

Keywords: Navigation, Learning, Memory

Navigation in daily life requires continuous egocentric experiences. How do we find our way home from the market so well? How do we bind the map in our mind to continuous upcoming egocentric experiences? To answer this question, we design a navigation experiment and evaluate how well humans retrieve past experiences.

Participants are asked to navigate a 5 X 5 maze while collecting diamonds one at a time as fast as possible over four pairs of conditions. In an egocentric training condition, participants control an avatar navigating the maze as a first-person perspective. As a result, they can only find a diamond by exploring. In an allocentric training condition, they control the avatar from top-down view so that they can identify the diamond and the maze structure simultaneously. In the egocentric test condition, the walls of the maze become invisible, and the participants need to navigate without colliding with the invisible walls. The same is true for the allocentric test condition. These two training and test conditions are combinatorially paired to construct the four pairs of conditions. In this presentation, we discuss to what extent participants rely on the first-person experiences in solving maze in the allocentric condition and vice versa.

This work is supported by RS-2023-00217361.

P-04

Event coding of subiculum neurons in virtual environments

Su-Min Lee, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Hippocampus, Subiculum, Episodic memory

The subiculum is located at the junction where hippocampal information is transmitted to the cortical and subcortical areas, but little is known about its functional significance. In our previous study (Lee et al., J Neurosci 2018), it was reported that a portion of subicular neurons organize their firing fields based on behavioral events rather than position, unlike the hippocampal place cells. Extending these findings, we hypothesize that the subiculum, which receives fine-tuned positional information from the hippocampus, would reconstruct it into behavioral resolution. We developed a virtual reality-based task for rats to verify such coding in the subiculum. In this task, a rat shuttled a circular track to obtain rewards (i.e., drops of honey water) while experiencing two virtual environments filled with different visual landmarks. The reward is always given at a fixed position (0°), but the start location is randomly selected from one of three positions (90°, 180°, 270°) for each trial. At the reward location, the rat was required to choose one of the two lick ports associated with the virtual environment it was running in to receive a reward. After choice, the rat needed to reach the start location again to end the trial. This task scheme enabled us to examine whether the firing fields of subicular neurons flexibly change based on the behavioral events (i.e., pre-choice vs. post-choice) even if the rat receives the same visual inputs. We are recording single-unit activities and local field potential in the hippocampus and subiculum simultaneously while rats perform the task, and we will show the preliminary data.

This research was supported by the National Research Foundation of Korea Grants 2018R1A4A1025616, 2019R1A2C2088799, 2021R1A4A2001803, 2022M3E5E8017723, 2023R1I1A1A0106975612 and the BK21 program.

P-05

Laminar-selective representation of textures in the mouse primary somatosensory cortex

Daehyun Oh, Jeehyun Kwag*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Somatosensory cortex, Tactile perception, Electrophysiology

Tactile perception, which relies on whisker, plays a pivotal role in distinguishing various texture elements among rodents. Previous studies have established that different textures elicit distinct spike patterns in neurons located within the laminar-structured primary somatosensory cortex (S1). However, how neuronal spikes in each layer represent different texture information is poorly understood.

To address this question, single-unit activities in S1 were recorded while presenting texture to the whiskers of the mouse. The majority of neurons in S1 exhibited distinctive spike patterns in response to these textures. To investigate whether individual cells in each layer were capable of differentiating between different textures, spike firing rates and timing were compared during texture stimulation. Our preliminary results demonstrated that each layer displayed a different fraction of single cells which exhibited discriminative capabilities for rough and smooth textures. Together, our preliminary results indicate that each layer in S1 may employ distinct features of spike patterns to encode texture information.

NRF-2019M3E5D2A01058328

P-06

Temporally heterogeneous coding of multisensory object information in the perirhinal cortex

Heung-Yeol Lim, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Perirhinal cortex, Object recognition, Multisensory

Our brain can easily integrate multiple sensory modalities to accomplish object recognition. The perirhinal cortex (PER) is the candidate region for this process since it is critical for object recognition and receives direct inputs from all sensory modalities. However, little is known about the neural mechanisms by which this region supports multisensory object recognition. We recorded single-unit spiking activities in the PER while rats performed a multisensory object recognition task. In this task, rats were trained to recognize two different multimodal objects, each associated with unique visual and auditory features (i.e., a boy image with 5kHz tone vs. an egg image with 10kHz tone). The rat sampled an object by poking their nose to the center hole. After sampling, the animal was required to choose either left or right reward port based on the object identity. Rats were trained with the multimodal stimuli (both visual and auditory cues co-present), but were also tested with unimodal stimulus once they learned the task. When the PER was inactivated with muscimol (MUS; n=6, male Long-Evans rats), there were significant performance deficits in all retrieval conditions (multimodal: PBS 85% and MUS, 70%; visual: PBS 73% and MUS 59%; auditory: PBS 74% and MUS 61%). Single-unit spiking activities were recorded from the PER in another group of rats (n=8). PER neurons showed transient object selectivity that temporally tiled the task epoch (i.e., from stimulus onset to choice response). We further examined the modality bias of the selectivity and classified neurons into visual (n=19), auditory (n=27), or unbiased (n=104) category. While most auditory neurons showed direct and strong responses to auditory information immediately after the sound onset, visual or unbiased neurons showed temporally distributed selectivity. These results suggest the importance of temporal coding in the PER to represent both the identity and modality information of a multisensory object.

This research was supported by the National Research Foundation of Korea Grants 2018R1A4A1025616, 2019R1A2C2088799, 2021R1A4A2001803, 2022M3E5E8017723, and the BK21 program.

P-07

Impairment of Prefrontal Neuronal Ensemble Activities during Memory Retrieval in the 5XFAD Mouse Model of Alzheimer's Disease

Yoonsoo Yeo, Jeehyun Kwag*

Dept. Brain and Cognitive Sciences, Seoul National University

Keywords: Prefrontal cortex, Alzheimer's disease, Long-term memory

Alzheimer's Disease (AD) is characterized by a progressive decline in neuronal function, resulting in impairments in memory consolidation and retrieval. The prefrontal cortex (PFC), which is responsible for long-term memory processes, is believed to be involved in this impairment. However, our understanding of how memory consolidation and retrieval are represented in groups of PFC neurons and their impact on memory deficits in AD is limited. In our study, we used Ca^{2+} imaging in both C57BL/6j mice and a 5XFAD mouse model of AD. We recorded the activity of PFC neurons before and after contextual fear conditioning. Our findings revealed distinct changes in the ensemble activity of excitatory neurons in C57BL/6j mice from day 1 to day 14. In contrast, 5XFAD mice showed impaired changes in ensemble activity across all sessions. These results suggest that alterations in ensemble activity in the PFC may contribute to the retrieval of long-term memory, and impaired changes in ensemble activity may contribute to memory deficits in AD.

P-08

Non-spatial neural firing patterns of hippocampal place cells are reactivated during visual scene memory task

Jae-Min Seol, Su-Min Lee, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Hippocampus, Sharp-wave ripple, Non-spatial, Memory, Consolidation

It is necessary for the hippocampus processing spatial and non-spatial information for episodic memory. For re-processing, representations of spatial information are reactivated during SWRs. But what about non-spatial information? To test whether non-spatial information area also represented in SWRs, we reanalyzed our previous data (Lee et al., 2022) recorded from (n=260) and subiculum (SUB; n=151) place cells while rats (n=5) doing a scene memory task in a T-maze, and defined place fields by spiking phases relative to theta oscillation. We then analyzed place cell groups co-activated in single SWRs (n=933 in CA1, n=1994 in SUB) during awake immobile period. We decoded spatial positions from spikes in SWRs and conducted linear regression, and detect significant replay by comparing regression coefficient to that of the shuffled data ($\alpha=0.05$). To find non-spatial reactivation, we measured rate remapping between scenes or responses for co-activated cells, and compared averaged rate remapping degree to subsampled data ($\alpha=0.05$). Among the ensembles (26.5% in CA1, 65.5% in SUB) significantly reactivated during SWRs, CA1 had more spatial replay (18.3%) than SUB (7.8%), while SUB had more non-spatial reactivation (46.9%) than CA1 (7%) (χ^2 test, $p<0.0001$). Some groups in CA1 (1.2%) and SUB (10.8%) had both types of reactivation. This shows that place cells also reactivate non-spatial information during SWRs, and SUB may be important for consolidating such task information.

This research was supported by the National Research Foundation of Korea Grants 2018R1A4A1025616, 2019R1A2C2088799, 2021R1A4A2001803, 2022M3E5E8017723, and the BK21 program.

P-09

Shared and distinct ACC neural circuits encoding affective pain and empathy

Jiye Choi, Jeeyeon Kim, Sungjoon Choi, Sehoon Keum*

Center for Cognition and Sociality, Institute for Basic Science

Keywords: Observational fear, Anterior cingulate cortex, Affective pain

Understanding the emotional states of others is essential for human social interactions. The anterior cingulate cortex (ACC) has been implicated as a central neural hub responsible for coordinating emotional and cognitive processes involved in empathy. However, how the ACC neurons represent the sharing of affective experiences in response to distress signals from conspecifics remains unclear. Here, using miniature endoscopic calcium imaging in freely behaving mice, we identify distinct and dynamic neural ensembles in the ACC that encode observational fear, a rodent model of affective empathy. Notably, decoding analysis demonstrates that neural activity encoding vicarious freezing is preserved in ACC excitatory neurons during consecutive observational fear and selectively recruited by emotional pain induced by foot shocks. We further demonstrate that optogenetic inhibition of the ACC to the periaqueductal gray (PAG) circuit impairs both pain affective-motivational behavior and observational fear, but not to the basolateral amygdala (BLA). Accordingly, distinctive activity patterns in PAG projecting ACC cells is represented during observational fear. Collectively, our findings establish a functional role for ACC neural representations in shaping observational fear through the encoding of affective pain.

This work was supported by Institute for Basic Science (IBS) grant IBS-R001-D2, Republic of Korea.

P-10

Dissociated Task-Specific Response in Hippocampal-Dependent Virtual Reality Environment-Based Working Memory Task in PL and IL

Hee-Seung Ha, Eun-Hye Park, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Medial prefrontal cortex, Working memory, Electrophysiology

A working memory(WM) task involves sequential task phases: An essential cue is acquired in the sample phase, maintained during the delay phase, and a decision is reached during the response stage. The medial prefrontal cortex(mPFC) plays critical roles in WM, but how information is processed at the single neuronal level during different task phases remains largely unknown, particularly within subregions, the prelimbic(PL) and infralimbic(IL) areas. In the study, 5 male Long-Evans rats were in a virtual reality WM task where the body-fixed rat ran on a cylindrical treadmill surrounded by 3 LCD monitors. The rat was exposed to one of four virtual environments during the sample phase. Next, the rat ran for 3 seconds in the delay phase. In the response phase, the rat had to choose between licking the left or right licking port, which corresponded to the sampled scene, to receive a liquid reward (120 trials/session). Inactivating the intermediate hippocampus(iHPC) with muscimol decreased performance compared to vehicle controls in the same rats ($t(11)=6.60$, $p<0.001$), indicating that the task required information transfer from the iHPC to mPFC. The activity of mPFC cells recorded from 4 of the same rats was modulated by visual environment and choice information, but iHPC inactivations significantly disrupted such modulatory roles of task components differentially across task phases and subregions ($p's < 0.05$; Wilcoxon rank-sum tests). Remarkably, the inactivation of the iHPC disrupted scene-association rate modulation only in the sample phase, both in PL and IL areas. iHPC inactivation leads to disrupted choice-associated rate modulation in the delay phase in PL and all 3 task phases in the IL area. Our results show PL area process information in a task phase-specific manner. In contrast, IL may be essential in ensuring that the information gathered from the iHPC is linked to the subject's decision throughout the WM task's different stages.

The National Research Foundation of Korea Grants 2022M3E5E8017723, and the BK21 program.

P-11

Medial prefrontal cortex subregions differentially mediate social stress-induced increases in generous behavior

Kun-il Kim, Min-Young Kim, Kyung-Hwan Lee, Hackjin Kim*

School of Psychology, Korea University

Keywords: Stress, mPFC, decision-making

It has been reported that social stress can have contradictory effects on social behavior. In this study, by incorporating stress treatment with a modified social discounting task (MSDT), we investigated the neural mechanism underlying the association between the physiological responses to social stressors and inter-subject variability in money-sharing behavior. After Montreal imaging stress task, samples of their saliva were collected to measure changes in cortisol concentration from the baseline. Then, participants performed a MSDT, which was designed to measure money sharing with targets of varying social distance when money sharing results in inequal or equal consequence. We used computational modeling and hierarchical Bayesian analysis to identify the three distinctive dimensions of motivation: 1)the sensitivity to target's social distance, 2)the sensitivity to reward for self, and 3)the degree of disadvantageous inequality aversion as measured by the tendency of avoiding an option that gives more money to a target than to self. A negative correlation was shown between the degree of increase in cortisol concentration and the degree of disadvantageous inequality aversion. The degree of increase in cortisol concentration was negatively correlated with the degree of the DMPFC activity in tracking the value of reward for self in the disadvantageous inequal condition. In turn, those with higher degree of disadvantageous inequality aversion exhibited higher degree of DMPFC activity in tracking the value of reward for target in the disadvantageous inequal condition. Finally, a mediation analysis confirmed that the association between the degree of increase in cortisol level and the degree of disadvantageous inequality aversion was indirectly mediated by the DMPFC activity. These suggest that the DMPFC plays a key role in promoting disadvantageous inequality aversion, which can be interrupted by socially induced stress, resulting in self-protective prosocial motivation.

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2021M3E5D2A0102249311).

P-12

Machine-learning-based behavior categorization of rat behaviors in the spontaneous object exploration paradigm

Jae-Woo Kim, Su-Min Lee, Seong-Beom Park, Eun-Young Lee, Inah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Hippocampus, Object exploration, Machine learning

The spontaneous object exploration (OE) paradigm is commonly used to measure rodent behavior in the hippocampus study. The duration of OE has been typically measured by the time an animal's snout stays within a certain distance from an object. However, this method is too simplistic and may overlook the diverse behaviors exhibited by rodents, and introduce subjective bias of the experimenter. To address this issue, we conducted a reanalysis of OE tasks in our previous study (Park et al., 2022) using Deeplabcut (DLC), a pose estimation software. We used video recordings of 23 male Long-Evans rats for this analysis. Preliminary examination was conducted on the 'novel object preference (NOP)' task, which consisted of three identical 5 min sample phases followed by one 5 min test phase after an 1 hr delay. In the sample phases, two identical objects were presented, with one object being replaced by a novel one in the test phase. DLC was trained with 15 points marked on the rats' body parts and the objects. A custom Matlab script was employed to classify 15 behavioral types of the rats. Among these types, we defined a behavior called 'active OE' based on specific criteria. Compared with the results of the previous analysis, the DCL analysis resulted in a shorter duration of active OE for the novel object during the test phase ($p < 0.01$; paired t-test). This was because the previous OE measurement included various object-related behaviors (stepping on, jumping over, passing by, etc.), which did not occur significantly more frequently for the novel object compared to the familiar one in the test phase. Importantly, the duration of rearing against the arena walls increased in the test phase compared to the sample phase ($p < 0.05$). Our findings suggest that rodents demonstrate diverse behaviors when exposed to novelty in their environment, and utilizing a machine-learning-based approach can help identify critical behaviors that have been overlooked by human observers before.

This research was supported by the National Research Foundation of Korea Grants

2018R1A4A1025616, 2019R1A2C2088799, 2021R1A4A2001803, 2022M3E5E8017723, and the BK21 program.

P-13

Neural mediators of the relationship between self-efficacy and self-protective tendency

Jinhee Kim, Hyeyoon Jung, Daon Lee, Hackjin Kim*

School of Psychology, Korea University

Keywords: Self-efficacy, self-protective bias, task-based fMRI

Self-efficacy is one's belief in their own ability to achieve their goals and is known to be a protective mechanism to preserve mental health. Empirical evidence suggests that self-efficacy serves as a buffer against other people's opinions, and its underlying neural mechanism remains unknown. Here, we used functional magnetic resonance imaging (fMRI) to scan the brains of 36 participants while they performed a reciprocal artwork evaluation task, involving receiving an evaluation of their own artwork from a partner and evaluating the same partner's artwork. Choice behavior influenced by evaluative feedback were fit with the reinforcement learning model to estimate the learning rates as well as trial-by-trial fluctuations in the self-protection value and prediction error (PE), which were mapped to fMRI data. We used the general self-efficacy scale to assess the individual's self-efficacy trait. The behavioral results revealed that those with lower self-efficacy showed higher learning rates. We found that the degree to which the right ventrolateral prefrontal cortex (VLPFC) encodes PE at the feedback receipt phase is associated with self-efficacy differences. Specifically, the VLPFC activity of individuals with high self-efficacy increased in response to unpredicted positive feedback whereas that of individuals with low self-efficacy increased in response to unpredicted negative feedback. Lastly, we found that the precuneus activity tracking the social feedback type during the partner's artwork evaluation mediates the relationship between self-efficacy and learning rate. Taken together, these findings suggest that the VLPFC and precuneus, which are known to be involved in emotional regulation and self-referential processing, track different aspects of social feedback depending on individual differences in self-efficacy, thereby providing a neural explanation for why individuals with lower self-efficacy show self-protective bias due to negative social feedback.

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2020R111A1A01070413) and a Korea University Grant.

P-14

Interplay between Temporal and Directional Expectations on Sensorimotor Processing

Sanghum Woo^{1,2}, Sua Bae¹, Jeongwon Kim¹, Gunwoo Kim¹, Yechan Hwa¹, Joonyeol Lee^{1*}

¹Center for Neuroscience Imaging Research, Institute for Basic Science / Department of Biomedical Engineering, Sungkyunkwan University, ²Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University

Keywords: Expectation, Bayesian observer model, Smooth pursuit eye movement

The ability of humans to make optimal decisions relies on their expectations concerning time and direction in the world, and previous research has shown the profound influence of each expectation on behavior. Higher temporal expectations yielded faster reaction times, while greater directional expectations led to biased behavior toward the expected direction and reduced variability. However, it remains unknown how their interaction exists and affects our behavior. This study aims to explore the existence of the interaction between these two expectations and propose a Bayesian observer model as the underlying mechanism. Thirty human participants are involved in a smooth pursuit eye movement task, while electroencephalography (EEG) and eye movement data were recorded simultaneously. We confirmed a close relationship between temporal and directional expectations through correlation analysis, which revealed a significant negative correlation between the reaction time of eye movements and directional bias. To explore a mechanism about the interaction, we developed two alternative hypotheses and employed the Bayesian observer model for both: (1) Temporal and directional expectations increase simultaneously, or (2) considering the limitations of our cognitive capacity, as more resources are allocated to temporal expectations, fewer resources remain for processing the direction of moving target, resulting in reduced reliability of sensory information and an increased reliance on directional expectations. Through the modeling, we determined which hypothesis provides a more accurate explanation for the bias and variability observed in the direction of movement. When using the Akaike's weight as a measure to evaluate the goodness of fit, we observed that most participants supported the first model. We anticipate that this model will provide valuable insights into the integration of expectancy information in the brain.

This research was supported by IBS-R015-D1 and the National Research Foundation of Korea grant funded by the Korean government (MSIT) (No. 2023R1A2C2005290).

P-15

Distinct Integration of Opposite Valence Across Multiple Brain Areas

Ji-Hye Yang, Hyung-Goo Kim*

Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Korea,
Center For Neuroscience Imaging Research, Institute for Basic Science, Suwon, Korea

Keywords: Striatal Dopamine, Salience, Associative Learning

Dopamine (DA) neurons play a crucial role in reward learning. In particular, DA circuits from the ventral tegmental area (VTA) to the ventral striatum (VS) have been intensively studied as their signals encode reward prediction errors (RPEs). However, some DA neurons projecting to other subregions are excited by aversive or alerting events, suggesting that these neurons may encode emotional salience caused by the stimuli. Salience is commonly conceptualized by ‘unsigned RPE’ or ‘absolute RPE’ but the computing process remains elusive. To examine the underlying process, we assumed that the complex of ambivalent inputs would be more salient. In case that ‘good surprise’ and ‘bad surprise’ come together, the salience would be greater. Based on the idea, we designed the experiment with the opposite valence simultaneously.

To address this, we trained mice for a visual classical conditioning task. Visual stimuli predicted one of the four outcomes: neutral, positive (water), negative (electric tail shock), and the combination of the opposite valence (water and electric shock together). We recorded extracellular DA levels across multiple subregions using fiber photometry. DA activity in the ventral striatum increased with the value of the outcome, corresponding to the RPE. Most animals exhibited inhibitory responses to the shock condition. Excitatory response in the combination condition was weaker than that in the water condition. These results are consistent with the RPE theory.

In contrast, dorsal striatum displayed an excitatory response to the shock condition and a stronger response to the combination condition, suggesting that dorsal striatal DA neurons encode salience instead of the RPEs. The results suggest that salience signals are internally integrated through the sum of each ‘unsigned RPE’. For future research, we will analyze and quantify the data to determine whether the effect of salience signals contribute to associative learning alongside RPE signals.

We would like to express our sincere gratitude to all my colleagues and staff at Sungkyunkwan University and the Center for Neuroscience Imaging Research.

P-16

Exploring the Implicit Dynamics of Hierarchical Sequence Learning

Minho Shin, Hyeon-Ae Jeon*

Department of Brain Sciences, Daegu Gyeongbuk Institute of Science and Technology

Keywords: implicit sequence learning, ideal observer model, Bayesian inference

Humans construct their world knowledge hierarchically to behave more flexibly and efficiently. Yet, little is known about how we implicitly construct internal representations of the environment in a hierarchical manner from a stream of information when no explicit feedback is given. To address this issue, we designed a sequence learning task in which participants passively responded to a continuous stream of stimuli. Specifically, we instructed 25 online participants to respond as quickly as possible to the given stimuli using the corresponding keys. The stimulus presentation was governed by a hierarchical structure that they were unaware of. This hierarchical structure consisted of a few hidden states, where a particular stimulus was predominantly presented or all stimuli were randomly presented. We hypothesized that participants would take advantage of a hidden structure, leading to faster response times. To examine human behaviors, we introduced several ideal observers. First, we modeled an ideal observer that optimally learns the hidden hierarchical structure using a hidden Markov model. We refer to this model as the hierarchical model. As an alternative, we selected the Dirichlet-categorical model. This observer predicts the next stimulus based on the number of previous first-order transitions between stimuli. As this model lacks any information concerning hidden states, it is termed the flat model. To compare the performance of each model with that of humans, we computed surprise, uncertainty, and belief update based on the models' predictions. These derived measures served as predictors for analyzing the participants' response time data using a mixed-effects model. Our results demonstrated that human behaviors were best explained by the hierarchical model that optimally learns the hidden hierarchical structure. Therefore, we suggest that people have the ability to construct their world knowledge hierarchically, even without awareness.

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) (NRF-2019M3C7A1031995, NRF-2020R1A2C2099568).

P-17

Reward processing in lateral habenula neurons using virtual reality track

Jongwon Yun, HyungGoo Kim*

Department of Biomedical Engineering, Sungkyunkwan University

Keywords: Lateral habenula, Negative temporal difference error, Electrophysiology

The ability to learn from past experiences and make decisions based on the expected value of rewards and punishments is essential for survival. The activity of dopamine neurons in the ventral tegmental (VTA) is thought to signal a reward prediction error, which is the difference between the expected and actual rewards. The lateral habenula (LHb) is a brain region that provides input to dopamine neurons and projects to other brain regions involved in emotion and motivation. LHb neurons increase firing when reward is omitted and decrease firing after receiving a reward. These activities are thought to signal an inverse of reward prediction error. Our previous study using virtual reality (VR) tracks showed that dopamine neurons encode moment-by-moment temporal difference (TD) error. Based on the fact that LHb neurons provide inhibitory signals to dopamine neurons, we hypothesized that LHb neurons encode negative TD error.

In this experiment, we used VR to study the activity of LHb neurons. We trained mice ($n = 5$) to run a VR track to earn water rewards. We then recorded the spiking activity of LHb neurons ($n = 259$) using Neuropixels while the mice ran the track and received rewards in various experimental conditions. In some trials, animals were suddenly moved to a position farther to the reward. We observed transient excitations to the backward teleportation. In contrast, we observed transient inhibition in response to forward teleportations that moved animals closer to the reward. These activities resembled the opposite characteristics of TD errors in dopamine neurons, a form of negative TD errors.

I would like to thank Mijoo Park, YunAh Lee, who worked with me on the neuropixels setup, and the NeuRLab team for their support.

P-18

Reinforcement Learning-based Rule Generalization: Distinguishing True and Pseudo-Generalizers in Sequential Rules

Taehyun Yoo, Hyeon-Ae Jeon*

Department of Brain Sciences, Daegu Gyeongbuk Institute of Science & Technology

Keywords: Rule generalization; Reinforcement Learning; Sequence

The ability to generalize rules in novel environments is a remarkable characteristic of human beings. While previous studies have explored their ability to generalize rules based on perceptual similarity between stimuli, little is known about how humans generalize sequential knowledge. In this study, we devised a sequence learning task wherein participants were asked to judge the correctness of structural dependencies based on visual features indicated by cues. Each cue consisted of shape and color, representing structural dependency and visual feature, respectively. Prior to the experiment, participants were explicitly instructed about the rules associated with three cues. Additionally, six novel cues were presented alongside the three instructed cues during the experiment.

To investigate the process of generalization, we developed four distinct models within a reinforcement learning (RL) framework. These models differed in their state-space, where states were defined by the mappings between cue and sequence that participants may track. Model 1 included states where cue shape and cue color mapped to structural dependency and visual feature, respectively. Model 2 and Model 3 contained states where cue shape and cue color mapped solely to structural dependency and visual feature, respectively. Model 4 included states where cue shape and cue color mapped to visual feature and structural dependency interchangeably.

Our findings revealed that participants whose winning models were model 1 successfully discovered rules showed superior generalization abilities compared to those whose winning models were others. We demonstrated that the best explanation for generalization performance was provided by the full model, which included states representing structural dependencies based on the corresponding visual features. These results shed light on the computational mechanisms underlying successful rule generalization through behavioral modeling.

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) (NRF-2019M3C7A1031995, NRF-2020R1A2C2099568).

P-19

Random imitation facilitates population survival optimization in multi-agent systems

Gwangsu Kim¹, Minsoo Kang¹, Hyeonsu Lee¹, Youngjin Park¹, Min Song^{1,2}, Se-Bum Paik^{3*}

¹Department of Bio and Brain Engineering, KAIST, ²Program of Brain and Cognitive Engineering, ³Department of Brain and Cognitive Sciences)

Keywords: Random imitation, Multi-agent simulation, Social learning

Imitation, the act of copying others' behaviors, is considered an essential behavior of humans and animals that facilitates learning and development. Conventional theories suggest that humans and animals should copy more advantageous traits as opposed to those chosen at random, as a payoff-biased strategy. However, it has been reported that imitation in humans and animals often takes the form of random copying, with the possibility of imitating a suboptimal target. This raises the question of whether random copying can be advantageous enough to serve as a successful behavioral strategy. Here, using a multi-agent survival model simulation, we show that random copying can optimize individual agent traits, leading to population-level survival optimization. In our model, the lifespan of the agent is determined by a single trait that can be modified through imitation or by random mutation. When agents were only subject to mutation, the average lifespan did not change over time. In contrast, when agents were allowed to randomly imitate each other, their average lifespan increased over time, and their traits converged to the optimal trait in the given environment. By modulating the timings and frequencies of random copying, we also found that early one-shot random copying maximizes the lifespan increase of the agent group. The beneficial effect of random copying was comparable to or even superior to that of selective copying, suggesting that random copying has the potential to serve as a better learning strategy in a multi-agent system. Further generalization of the model to multi-trait settings confirmed that the optimization effect of random copying becomes more pronounced as the number of traits increases. Moreover, we found that the key characteristics of random copying are indeed observed in animal behavior data. Our findings highlight that random copying can be an effective behavioral strategy for population-level optimization in a multi-agent system.

This work was supported by Nos. NRF-2022R1A2C3008991, NRF-2021M3E5D2A01019544, NRF-2019M3E5D2A01058328 (to S.P.).

P-20

Neurobehavioral impacts of a social observer on risky decision-making in cigarette smokers

HeeYoung Seon¹, Cheolin Yoo², Dongil Chung^{1*}

¹Department of Biomedical Engineering, Ulsan National Institution of Science and Technology (UNIST), ²Department of Occupational and Environmental Medicine, Ulsan University Hospital, University of Ulsan College of Medicine

Keywords: Smoking, Social influence, Risky decision-making

Smoking is a social behavior. Being around other smokers may induce earlier initiation of smoking, but at the same time, help from a social support group is often suggested to be a way to stay abstinent. Although it is known that individuals have a tendency to follow preferences of others who are around, it has not been examined how smokers, individuals who are known to have biased preferences for risky behaviors, would respond to such social environments. Smokers may have exaggerated susceptibility to risky social information, and thus be more influenced by the presence of risky others. Alternatively, smokers may identify themselves more similar to risky others, and face no need to alter their choices to be riskier. To test these hypotheses, we examined risky decision-making in smokers, non-, and past- smokers using a gambling task. On some trials, participants were informed that their choices would be observed by a randomly matched risk-averse or risk-seeking partner. After learning about the observers' preferences, smokers, unlike non- or past- smokers, responded that they believed to be more similar with the risky observer than the safe observer. During the task, non- and past- smokers made riskier (or safer) choices when they were observed by a risky (or safe) observer, while smokers' choices under the observation of a risky observer were comparable to the choices they made alone. Consistent with this behavioral pattern, smokers showed different event-related potential (ERP) patterns from the other two groups. Specifically, past- and non- smokers showed larger N400, an ERP associated with semantic processes, in social compared to non-social conditions, whereas the same ERP differences were not present in smokers between the risk seeking observer versus non-social conditions. Our results suggest that smokers identify themselves risky and provide neurobehavioral evidence for biased impacts of a risky environment on smokers' decision-making processes.

This work was supported in part by the National Research Foundation of Korea (NRF-2018R1D1A1B07043582 to D.C.) and UNIST internal funding (1.230016.01 to D.C.). We thank Hyeji Lee, Jiwon Park and Minho Hwang for their research support.

P-21

Spontaneous emergence of face detection in untrained deep neural networks

Seungdae Baek¹, Min Song², Jaeson Jang¹, Gwangsu Kim³, Se-Bum Paik^{4*}

¹Department of Bio and Brain Engineering, Korea Advanced Institute of Science and Technology, ²Department of Bio and Brain Engineering, Program of Brain and Cognitive Engineering, Korea Advanced Institute of Science and Technology, ³Department of Brain and Cognitive Sciences, Life Science Research Institute, Korea Advanced Institute of Science and Technology, ⁴Department of Brain and Cognitive Sciences, Korea Advanced Institute of Science and Technology

Keywords: Face detection, Spontaneous emergence, Deep neural networks

Face-selective neurons are observed in the primate visual cortex and are considered the basis of face detection in the brain. This observation has led to crucial inquiries on the developmental mechanisms of these neurons: whether these neuronal tunings can emerge innately or require visual experience, and whether face tuning represents a unique function distinct from other visual objects. Recently, researchers have utilized biologically-inspired deep neural networks (DNNs) to simulate and predict the responses of the ventral visual stream to gain insights into the mechanisms of brain functions. Here, using DNN model of ventral visual stream, AlexNet, we propose a mechanism that face-selectivity can arise without any training. By measuring responses of the last convolutional layer to faces and non-face stimuli, we found face-selective units that show significantly higher responses to faces than non-face images. These units display face-selectivity indices comparable to those observed in the brain. Also, the emergence of face-selectivity is robust across diverse network randomization conditions. Next, to understand feature-selective response of face units, we reconstruct preferred feature images of each unit using reverse-correlation and generative adversarial network algorithms. The obtained images reveal that face units exhibit selectivity for face-like configurations, distinguishing them from non-selective units. Moreover, we confirmed that innate face-selectivity can perform face detection tasks. Intriguingly, we observe that untrained networks also exhibit innate selectivity to various non-face objects, challenging the notion of face-selectivity as a distinct form of tuning. This implies that selectivity to different objects can spontaneously arise in untrained networks. Our findings suggest that the initial random feedforward connections in untrained networks may be sufficient to initialize primitive face-selectivity and general selectivity to visual objects.

This work was supported by the National Research Foundation of Korea (NRF) grants funded by the Korean government (MSIT) (Nos. NRF-2019R1A2C4069863, NRF-2019M3E5D2A01058328, NRF-2021M3E5D2A01019544) and the Singularity Professor Research Project of KAIST (to S.P.).

P-22

Two-step value normalization accounts for the impacts of choice-unavailability on decision processes: an eye-tracking study

Minho Hwang, Dongil Chung*

Department of Biomedical Engineering, Ulsan National Institution of Science and Technology

Keywords: Value normalization; Decision-making; Attention; Eye-tracking; Computational modeling

Choices are often influenced by the dynamic context that changes over time. For instance, changes in an item's availability (e.g., an item being sold-out) affect subsequent purchasing choices. Previously, we suggested a Two-step value normalization model that provides a potential explanation for the dynamic value normalization process along with the changes in available options. Here, we collected eye-tracking data while individuals were making a series of gamble choices, and directly investigated whether or not the decision-makers took into account the value information of the gamble option even after the revelation of its unavailability. During the task, participants were provided with three gamble options where each option had a unique value. On some trials, one of the three options became unavailable ('sold-out') before participants were asked to make their decision, leading them to choose from the remaining two options. Consistent with our previous study, the Two-step value normalization model provided a better explanation for participants' decision patterns compared to other single step normalization models. Our eye-tracking data revealed that participants indeed briefly fixated their eyes on the sold-out option after the revelation of its unavailability. Moreover, the extent to which individuals fixated on the sold-out option was negatively associated with the individually estimated normalization parameter accounting for the normalization sensitivity to the sum of option values. This counterintuitive finding suggests a possibility that attention towards the sold-out option may contribute to an active inhibition process, which attenuates the inclusion of the unavailable option's information in the choice process. In sum, the current study provides physiological evidence for our suggested Two-step value normalization model.

This work was supported in part by the National Research Foundation of Korea (NRF-2018R1D1A1B07043582 to D.C.) and UNIST internal funding (1.230016.01 to D.C.).

P-23

The Role of the Hippocampus in Forecasting Future Events during Goal-Directed Behavior

Jiwoong Park, Won Mok Shim*

Center for Neuroscience Imaging Research, IBS / Department of Biomedical Engineering, Sungkyunkwan University /
Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University

Keywords: Hippocampal preplay, Functional MRI, Spatial navigation

The hippocampus has been reported to exhibit sequential neural activities known as hippocampal "preplay" during episodic experiences, involving the anticipation of future events that have not yet occurred (Dragoi and Tonegawa, 2011). In this study, we aimed to investigate the hippocampal activity preceding predictable events in the human brain using functional magnetic resonance imaging (fMRI) while participants engaged in a Minecraft-based interactive 3D video game. We designed a herding task, which required participants to interact with biological entities in a complex environment. The task involved hierarchical action planning and execution to successfully herd a flock of sheep to a designated area while avoiding the loss of sheep by preventing them from falling into puddles or being preyed upon by a fox. We hypothesized that increased hippocampal activity would be observed during the anticipation of predictable events, particularly when the potential loss of the reward was anticipated.

Our results demonstrated increased hippocampal activity preceding predictable events, such as sheep falling into puddles compared to unpredictable events, such as a fox appearing randomly during gameplay. Notably, this predictive hippocampal activity was not observed during participants' initial encounters of the corresponding event but became prominent with repeated exposure. In summary, our findings provide suggestive evidence of predictive hippocampal activity in the human brain during goal-directed behavior, specifically in the anticipation of predictable episodic events. These results suggest a potential link to hippocampal preplay observed in spatial navigation studies with rodents. Overall, our study highlights the crucial role of the hippocampus in forecasting and simulating the future events, particularly those associated with future rewards.

This work was supported by the Institute for Basic Science Grant (IBS-R015-D1), the Fourth Stage of Brain Korea 21 Project in Dept. of IPHC, SKKU, and National Research Foundation of Korea(NRF-2019M3E5D2A01060299 and NRF-2019R1A2C1085566).

P-24

Adaptive generalization and efficient learning under uncertainty

Jiwon Park, Dongil Chung*

Department of Biomedical Engineering, Ulsan National Institute of Science and Technology

Keywords: Adaptive generalization, Probabilistic reversal learning, Computational modeling

People often use recognizable features to infer the value of novel consumables. This generalization strategy is known to be beneficial in stable environments, such that individuals can use learned rules and values in exploring new situations. However, it remains unclear whether and how individuals adjust their generalization strategy in volatile environments where previously learned information becomes obsolete. We hypothesized that individuals adaptively use generalization by updating their beliefs about the credibility of the feature-defined value model. To test this hypothesis, we developed a novel probabilistic learning task where participants learn about the reward contingencies assigned to a set of stimuli. Critically, each stimulus was generated as a Gabor patch that has a unique orientation, and to covertly inform a monotonic association between the stimuli orientations and the assigned reward probabilities, participants were initially trained on three anchoring pairs (0°-0%, 45°-50%, and 90°-100%). Subsequently, participants had to learn about four new stimuli and were informed that reward contingencies could change during the task. Participants performed better when the novel environment remained consistent with the previously learned monotonic association between feature and reward, suggesting efficient utilization of prior knowledge. Against other accounts, we found that individuals incorporated an arbitration between feature-based reward generalization and model-based learning based on volatility tracking. Notably, our suggested model captured differential impacts of generalization dependent on the context-volatility, such that individuals who were biased the most toward generalization showed the lowest learning error in a stable environment, but the highest error in a volatile environment. This work provides novel insights into the adaptive usage of generalization and highlights the potential adverse effects of over-generalization in volatile contexts.

본 연구는 한국연구재단의 지원을 받아 수행된 연구임(NRF-2020S1A3A2A02097375)

P-25

Surprise elicits neural event segmentation in naturalistic social contexts

Jisu Ro¹, Ye Eun Seo², Luke Chang³, Won Mok Shim^{4*}

¹Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, ²Department of Psychology, Sogang University, ³Department of Psychology and Brain Science, Dartmouth College, ⁴Department of Biomedical Engineering, Sungkyunkwan University

Keywords: Surprise, Naturalistic social interaction, Event segmentation

Unexpected behaviors of others trigger updates in our beliefs about their traits, which in return enable us to make more precise predictions about their future actions. While previous studies have demonstrated the role of surprise signals in facilitating learning and discretizing continuous experiences, which involve abrupt changes in neural states, the neural mechanism underlying surprise-based learning in naturalistic social cognition remains poorly understood. This study aimed to identify the behavioral and neural signatures associated with surprise, which elicits belief updates, during the observation of naturalistic social interactions.

We hypothesized that surprising actions would induce changes in the representation of character traits and corresponding neural activity patterns, resulting in distinct neural event boundaries. To test this hypothesis, we collected fMRI data while participants watched a movie featuring a two-person conversation. Using the GSBS (Greedy State Boundary Search; Geerligs et al., 2021) algorithm, a data-driven method for detecting neural state boundaries, we examined the time-points at which shifts in neural states occurred. Separate groups of participants provided surprise ratings for the social interactions presented in the movie and assessed the perceived personality of characters along six trait dimensions after each scene.

Our results revealed a significant correlation between surprise ratings and changes in the representation of character traits. Furthermore, we found that neural event boundaries followed peaks in surprise ratings, particularly in brain regions involved in representing social information, such as the angular gyrus (AG) and posterior cingulate cortex (PCC). These findings suggest that surprise signals triggered by social actions of others induce shifts in neural states within the social brain regions, reflecting updated beliefs about others.

This work was supported by the Institute for Basic Science Grant (IBS-R015-D1) and National Research Foundation of Korea(NRF)(NRF-2019M3E5D2A01060299 and NRF-2019R1A2C1085566).

P-26

Computational modeling of learning mechanisms and decision biases in finite sequential decision-making

ChangHyun Lim, Oh-Sang Kwon, Dongil Chung*

Department of Biomedical Engineering, UNIST

Keywords: Learning ; Sequential decision-making; Computational modeling

Individuals' choices are made subjectively. As an example of real-life decision problems, finite sequential decisions are also known to deviate from optimal choices. Although our recent study suggested that such decision biases are accounted for by assuming subjectivity in computing optimal choice solutions, it has not been examined the process how such value functions are shaped differently in each individual. To address this issue, we collected behavioral data on an online version of our secretary problem from individuals who have broader demographic backgrounds. During the task, participants made choices to accept or reject a presented random number. Acceptance led to the next round, while rejection gave individuals up to four additional chances to evaluate new random numbers. We constructed a computational model that incorporates a learning strategy similar to model-based learning, which simulates updating of individuals' beliefs about the task structure and sets up decision criteria using dynamic programming, and a model-free learning strategy, which simulates tracking of cached values from previously observed stimuli. In addition, we introduced another value-based component to capture individuals' impulsivity, accounting for their systematic biases to accept more at earlier opportunities. With a formal model comparison against alternative nested models, our suggested model best explains participants' behavior and their changes across the task. Furthermore, estimated impulsivity-related parameters were positively correlated with individuals' impulsivity trait measures, suggesting that the model's bias component indeed explains individuals' trait-like characteristics. These results provide a computational account of how individuals learn from their experiences that may explain heuristic biases manifested in their finite sequential decision-making.

This work was supported in part by the National Research Foundation of Korea (NRF-2018R1D1A1B07043582 to D.C.) and UNIST internal funding (1.230016.01 to D.C.).

P-27

Exploring spatial regularity of rewards in a naturalistic environment

Jeon Jae Young¹, Seng Bum Michael Yoo^{1,2,3*}, Won Mok Shim^{1,2,3*}

¹Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University,

²Center for Neuroscience Imaging Research, IBS, ³Department of Biomedical Engineering, Sungkyunkwan University

Keywords: Foraging, Bayesian inference model, 3D naturalistic task

Understanding the spatial regularity of rewards in a given environment is crucial for efficient foraging within partially observable naturalistic settings. While previous studies on patch leaving have successfully uncovered cognitive principles of successive economic decision-making, abstracting out spatial information into reward quantity and time cost domains may lead to the loss of essential components of the real world, given that foraging is an ecologically relevant behavior. Here we introduce an agent-based model that infers the spatial regularity of the given map through complex sequential Bayesian hypothesis testing. From the modeling exercise, we found that 1) humans are sensitive to the spatial pattern beyond the simple non-spatial information (e.g., reward amount), 2) humans chose upcoming action to maximize the information (i.e., active sampling) that verifies their current belief about the regularity of the map. In subsequent foraging, spatial patterns of rewards were actively exploited to improve foraging outcomes. Furthermore, we are exploring how the human brain represents dynamically updated information on spatial regularities. Overall, we claim that spatial information should be seriously considered when studying foraging, as it can bridge neuroscience with ecologically valid behavior.

This work was supported by Institute for Basic Science Grant (IBS-R015-D1), National Research Foundation of Korea(NRF) (NRF-2019M3E5D2A01060299 and NRF2019R1A2C1085566), Fourth Stage of Brain Korea 21 Project in Department of Intelligent Precision Healthcare(S-2023-0794-000), Sungkyunkwan University (SKKU)

P-28

Value and spatial information processing in the hippocampal-retrosplenial neural circuit

Daeul Kong, Min Whan Jung*

Department of Biological Sciences, Korea Advanced Institute of Science and Technology; Center for Synaptic Brain Dysfunctions, Institute for Basic Science

Keywords: Retrosplenial cortex, Hippocampal-cortical interaction, Value-based decision making

Through experiences, brain can construct an internal spatial representation of an environment ('cognitive map') by integrating information about spatial location and various non-spatial features. In a dynamically changing environment, it is essential to represent spatial information with combining expected outcome, that is value, in order to make an optimal choice. However, it is still poorly understood how the brain integrates value and spatial information. Previous studies revealed that the hippocampus (HPC) and retrosplenial cortex (RSC) are involved in value-based decision making and have important roles in spatial representation. Furthermore, they are anatomically connected and physiologically relevant each other. These suggest that hippocampal-retrosplenial circuit might integrate and process value and spatial information to guide optimal spatial navigation. Therefore, we focused on hippocampal-retrosplenial circuit to investigate how value and spatial information are processed and integrated in the brain. First, in vivo one-photon calcium imaging of RSC neurons has already been successfully conducted in mice performing a dynamic foraging task. There are various types of neurons which seem to represent choice, reward, value, and place, indicating that RSC may process both value and spatial information during the task. The next step involves simultaneous in vivo electrophysiological recordings in both the HPC and RSC, allowing for identifying activity patterns and correlations within the hippocampal-retrosplenial circuit. In addition, contributions of hippocampal inputs to value and spatial processing of RSC will be examined by observing changes in neural activities in RSC while modulating hippocampal activities. In conclusion, our study will uncover a neural mechanism for a cognitive map which is integrated with value. By elucidating these processes, a deeper understanding of how the brain represents and combines value and spatial information can be achieved.

This work was supported by the Research Center Program of the Institute for Basic Science (IBS-R002-A1; M.W.J.).

P-29

Adaptive goal-switching and planning for sequential decision-making in dynamic environments

Sang-Hyun Park¹, Dongil Chung², Won-Mok Shim^{1*}

¹Intelligent Precision Healthcare Convergence, Sungkyunkwan University, ²Department of Biomedical Engineering, UNIST

Keywords: adaptive target switch, plan diversity, sequential decision making, dynamic environment

Adapting plans and goals is crucial for maximizing rewards in dynamic environments. However, previous studies focused on simple decision-making tasks with fixed goals, leaving it unclear how humans adaptively switch goals based on value comparison. This study investigates the effects of diverse plans and adaptive goal-switching on decision-making performance. To simulate human behavior, we developed computational models that generate plans and evaluate target values. Subjects completed a calculation task, formulating equations to reach one of two target numbers. In the planning stage, subject selected a target number and formulated equations while plan diversity was measured by the proportion of components used. In the execution stage, subjects selected numbers and operators to complete the equation for the chosen target, with the option to switch targets. At each selection, unselected numbers could change randomly. Performance was assessed by the difference between the selected target and computed results. The planning and execution models utilized decision tree search algorithms, with parameters reflecting human behavior. The planning model predicted subjects' plans, while the execution model predicted selected numbers, operators, and goal-switching. The results showed a strong correlation between task performance and plan diversity. Faster calculations and accurate monitoring of the disadvantage associated with the current target, as estimated by the model parameters, were also correlated with improved performance. Additionally, preliminary fMRI results suggested that the anterior cingulate cortex and dorsolateral prefrontal cortex may play a role in representing the relative value of the current target compared to the alternative, particularly in subjects showing superior performance. In sum, our results highlight the significance of generating diverse plans and flexibly switching goals for effective decision-making in dynamic environments with multiple goals.

This work was supported by Institute for Basic Science Grant (IBS-R015-D1), National Research Foundation of Korea(NRF) (NRF-2019M3E5D2A01060299 and NRF2019R1A2C1085566), Fourth Stage of Brain Korea 21 Project in Department of Intelligent Precision Healthcare(S-2023-0794-000), Sungkyunkwan University (SKKU).

P-30

Role of dopamine in value processing in the entorhinal cortex-hippocampal system

Jaehun Kim¹, Minyeong Seo¹, Min Whan Jung^{1,2*}

¹Department of Biological Sciences, Korea Advanced Institute of Science and Technology, ²Center for Synaptic Brain Dysfunctions, Institute for Basic Science

Keywords: Entorhinal cortex-hippocampal system, Value processing, Dopamine

Representing values of different options is a fundamental component in making optimal choices for maximizing future rewards, and thus critical for survival in most animals. Recently, several studies found that the hippocampus might play a key role in value-based decision making. The entorhinal cortex (EC) is the gateway for information entering and leaving the hippocampal formation. EC superficial layers receive convergent cortical information, which is relayed to structures in the hippocampus, and hippocampal output reaches deep layers of EC, that project back to the cortex. Recent studies found that the medial entorhinal cortex (MEC) encodes reward information, which is deeply related to value. However, it is not known whether and how the MEC encodes value information, and affects hippocampal value processing.

Dopamine signals reward value and reward prediction error. The MEC receives dopaminergic inputs from the ventral tegmental area (VTA) and substantia nigra pars compacta (SNc) and contains dopamine receptors. The ultimate goal of this research is elucidating neural processes underlying dopaminergic regulation of value processing in the entorhinal cortex-hippocampal system. To this end, we first investigated the distribution of dopamine receptors in the superficial layers of the MEC in mice. In addition, our preliminary behavioral results from the probabilistic conditioning task with pharmacological inhibition of dopamine receptors in the MEC suggest that the entorhinal dopaminergic system may play an important role in value processing. We plan to characterize the release dynamics of dopamine in the entorhinal cortex and cell-type specific entorhinal value representations with dopaminergic regulation of them.

This work was supported by the Research Center Program of the Institute for Basic Science (IBS-R002-A1; M.W.J.).

P-31

Measuring impulsivity using a real-time driving task and inverse reinforcement learning

Sang Ho Lee¹, Myeong Seop Song¹, Min-hwan Oh², Woo-Young Ahn^{1*}

¹Department of Psychology, Seoul National University, ²Graduate School of Data Science, Seoul National University

Keywords: Impulsivity, Inverse reinforcement learning, Deep learning

Impulsivity has been extensively studied in relation to mental disorders and maladaptive behaviors using self-report questionnaires and behavioral tasks. A persistent issue is that self-report and behavioral measures show weak correlations between each other, although they are supposed to tap the same construct. To address this problem, we devised a real-time driving task called the “highway task” that allows participants to exhibit impulsive behaviors, such as reckless driving, which may mirror real-life impulsive traits assessed by self-report questionnaires. We hypothesized that the highway task would provide impulsivity measures that are strongly correlated with self-report measures of impulsivity. As hypothesized, statistical evidence supported the correlation between the performance in the highway task and a self-report measure of impulsivity (i.e., the Barratt impulsiveness scale, $r=0.46$). By contrast, measures of impulsivity from two traditional laboratory tasks (delay discounting and go/no-go tasks) did not correlate with BIS ($r=0.01, 0.07$, respectively). To infer subjective reward functions that underlie observed real-time behaviors in the highway task, we used an inverse reinforcement learning (IRL) algorithm combined with deep neural networks. The agents trained by IRL produced actions that resemble participants’ behaviors observed in the highway task. IRL inferred sensible reward functions from participants’ behaviors and revealed real-time changes in rewards around salient events (e.g., overtaking, a collision with a car ahead, etc.). The rewards inferred by IRL suggested that impulsive participants have high subjective reward values for irrational or risky behaviors. Overall, our results suggested that using real-time tasks with IRL may bridge the gap between self-report and behavioral measures of impulsivity, with IRL being a practical modeling framework for multidimensional data from real-time tasks.

This work was supported by the National Research Foundation grant funded by the Ministry of Science, Information and Communication Technologies and Future Planning, the Korean Government (Grant No. 2021M3E5D2A0102249311 [to W-YA]).

P-32

Roles of hippocampal interneuron subtypes in value processing

Jae Song Ju¹, Min Whan Jung^{1,2*}

¹Department of Biological Sciences, Korea Advanced Institute of Science and Technology,

²Center for Synaptic Brain Dysfunctions, Institute for Basic Science

Keywords: Hippocampus, Interneuron, Value processing

Evaluating the values of available options and choosing the best one is advantageous for survival. Although the hippocampus (HPC) is a key region of learning and memory and accumulating evidence showed the dependence of hippocampal activities by reward, the role in value processing and strong value signal in the CA1 region are recently identified. And the mechanism of hippocampal value processing is still unknown. CA1 consists of various types of GABAergic interneurons which play crucial roles in hippocampal functioning. Innervation of CA1 interneurons by dopamine and distinct reward modulation of their subtypes suggest important and distinct roles of interneuron subtypes in hippocampal value processing.

To investigate the roles of interneuron subtypes in hippocampal value processing, we recorded interneuronal activities in the dorsal CA1 region of head-fixed mice while mice perform a probabilistic classical conditioning task. Three different odor cues (CS: conditioned stimulus) were associated with different probabilities (75%, 25%, 0%) of 3 μ l water reward (US: unconditioned stimulus). One photon calcium imaging was used to target Parvalbumin (PV), Somatostatin (SST), and Vasoactive intestinal peptide (VIP) expressing interneurons which comprise a majority of the CA1 interneuron. Our preliminary data imply different dynamics of value-related activities in interneuron subtypes. We plan to confirm the roles of interneuron types in value-dependent behavior and value representation of pyramidal neurons. And identifying the roles of dopamine with the engagement of interneuron subtypes in CA1 value representation will further extend our understanding of the mechanisms of hippocampal value processing.

This work was supported by the Research Center Program of the Institute for Basic Science (IBS-R002-A1; M.W.J.).

P-33

Multidimensional risk profiles of impulsive decision-making for substance use disorders: similarities and differences between stimulants and opiates

Hyeonjin Kim¹, Elena Psederska², Jasmin Vassileva³, Woo-Young Ahn^{1*}

¹Department of Psychology, Seoul National University, ²Department of Cognitive Science and Psychology, New Bulgarian University, ³Department of Psychiatry, Virginia Commonwealth University

Keywords: Decision-making, Computational cognitive modeling, Substance use disorder

Impulsive decision-making is a multidimensional risk factor for substance use disorders (SUD). However, understanding decision-making differences across substances and their potential as endophenotypes for SUD is limited. We constructed a risk profile by using decision-making tasks (Iowa Gambling Task; IGT, Cambridge Gambling Task; CGT), Delayed Discounting Task; DDT, Balloon Analog Risk Task; BART) and compared the decision-making profiles between groups (heroin: N=157, amphetamine: N=140 SUD individuals; N=179 healthy controls; N=67 unaffected siblings of heroin; N=52 unaffected siblings of amphetamine) using a sibling comparison design. Through hierarchical Bayesian analysis, we computationally modeled task behavior and identified latent decision-making parameters. Participants were dependent on a single substance and had maintained abstinence. Euclidean distances between groups were also calculated to assess similarities and differences. Findings revealed shared risk factors in heroin and amphetamine groups, including deterministic betting on CGT and quick forgetting on IGT, observed in unaffected siblings too. Heroin group had unique risk factors: higher delay aversion on CGT, lower sensitivity/learning rates for losses on CGT/IGT, and higher preference for recent options on IGT. Unaffected siblings also showed lower sensitivity/learning rates for losses. Amphetamine group exhibited a unique risk factor of selecting options more randomly on DDT. Heroin and amphetamine groups had closest proximity, followed by siblings of heroin users and siblings of amphetamine users. Healthy controls had the greatest distance, followed by siblings of heroin users, siblings of amphetamine users, heroin users, and amphetamine users. Multidimensional impulsive decision-making risk profiles may serve as endophenotypes for stimulant and opioid use disorders, highlighting shared and unique characteristics among substances.

This work was supported by the Research Project Grant (R01DA021421) funded by the National Institutes of Health and the Creative Pioneering Researchers Program through Seoul National University to W.-Y.A.

P-34

Roles of intratelencephalic and pyramidal tract neurons of the prefrontal cortex in probabilistic reversal learning

Young Ju Yoon¹, Jee Hyun Yi², Sunghoon Jung¹, Min Whan Jung^{1*}

¹Department of Biological Sciences, Korea Advanced Institute of Science and Technology; Center for Synaptic Brain Dysfunctions, Institute for Basic Science, ²Center for Synaptic Brain Dysfunctions, Institute for Basic Science

Keywords: Prefrontal cortex, Classical conditioning, Behavioral flexibility

The prefrontal cortex (PFC) is known to play a crucial role in flexible control of behavior. To investigate roles of intratelencephalic (IT) and pyramidal tract (PT) neurons of the PFC in flexible control of behavior, we performed optogenetic modulation and calcium imaging of these neurons in deep layers of the medial PFC (mPFC) in adult mice engaged in a probabilistic classical conditioning task with cue-outcome contingency reversal. Our preliminary findings suggest that the effect of optogenetic modulation on behavior and neural activity dynamics during reversal learning differ between IT and PT neurons. Optogenetic inhibition of PT neurons, but not IT neurons, tended to increase the number of trials to reach the reversal criterion. Calcium imaging revealed that many PT neurons maintained cue- and expected outcome-dependent activity, but the majority of IT neurons displayed complex activity dynamics during the progress of behavioral reversal. Also, only the PT neuronal population showed an abrupt change in punishment-related responses following reversal onset. Although further studies are needed to determine specific functional roles of these cell types in probabilistic reversal learning, our findings so far suggest a more direct involvement of PT than IT neurons.

This work was supported by the Research Center Program of the Institute for Basic Science (IBS-R002-A1; M.W.J.).

P-35

Neural correlates of goal-directed control in internet gaming disorder and alcohol use disorder

Mina Kwon¹, Hang-Nyoung Choi², Harhim Park¹, Woo-Young Ahn¹, Young-Chul Jung²,
Woo-Young Ahn^{1*}, Young-Chul Jung^{2*}

¹Department of Psychology, Seoul National University, ²Department of Psychiatry, Yonsei University College of Medicine

Keywords: Internet gaming disorder; model-based decision-making system; computational modeling

Background: An imbalance between goal-directed (model-based) and habitual (model-free) decision-making systems is a common feature in addictive disorders. However, little is known about whether similar decision-making deficits appear in internet gaming disorder (IGD). This study compared neurobehavioral features associated with model-based and model-free systems in IGD and alcohol use disorder (AUD).

Method: Inside the functional magnetic resonance imaging (fMRI) scanner, individuals diagnosed as IGD (n = 22) and AUD (n = 22) and healthy control (HC; n = 30) performed the two-step task, a well-validated task for assessing the contribution of model-based and model-free systems in decision-making. We provided a mechanistic account of choice behavior using computational modeling and hierarchical Bayesian analysis. Then, we performed a model-based fMRI analysis to identify neural correlates of model-based and model-free prediction errors in each group.

Results: Computational modeling results suggest that the model-based weights did not differ among the three groups. However, neural correlates of the model-based prediction error dissociated the IGD group from the others. The IGD group showed significantly lower activation in the right orbitofrontal cortex compared to the AUD group ($t=-4.82$, $p<0.001$) and greater activation in the bilateral insula compared to the HC group ($t=4.66$, $p<0.001$).

Conclusions: We found distinct neural mechanisms underlying goal-directed behaviors in IGD, compared to AUD and HC. This finding suggests that excessive gaming might entail aberrant decision-making processes.

National Research Foundation (NRF) of Korea, Creative-Pioneering Researchers Program through Seoul National University.

P-36

A study on the effect of astrocyte-mediated CA3-CA1 synapse tuning on spatial firing and reactivation of CA1

Injae Yun¹, Min Whan Jung^{1,2*}

¹Department of Biological Sciences, Korea Advanced Institute of Science and Technology, ²Center for Synaptic Brain Dysfunctions, Institute for Basic Science

Keywords: Hippocampus, Astrocyte, Computational neuroscience

It is known that the hippocampus mainly represents spatial representations. Among the substructure, CA3 and CA1 are known to perform place coding, and it is known that CA1 performs more remapping by external stimuli such as reward site change and novel environment exposure compared to CA3. CA3 transmits information related to learning and memory to CA1 through the Schaffer collateral.

The rate of CA3 to CA1 excitatory synapse turnover increase under environment enriched conditions, and astrocytes have been shown to have a major role in eliminating synapses. It was confirmed that synapse down-regulation by astrocytic phagocytosis was inhibited in Megf10 K/O model mouse. It was confirmed that defects in synaptic plasticity and memory formation occur in a model mouse (loxP-floxed Megf10(Megf10^{fl/fl})) in which Megf10 can be conditionally deleted in a Cre-dependent manner. (Lee et al., 2021).

In this study, we plan to check differences in CA1 neuronal features between Megf10 K/O group and control groups. CA1 neurons have characteristics such as stable spatial firing, remapping in response to environmental changes, and sequential reactivation. For this purpose, we measure neural activity using calcium imaging. Mice were repeatedly exposed to linear or 2d maze, reward site change, cue rotation, novel maze exposure were planned for comparing remapping between experimental and control group.

Unlike simple CA3 inactivation, this experiment attempts to confirm the results of disrupting the regulation of CA3-CA1 synapse by astrocytes. This preserves the flow of information to CA3 to CA1, but prevents selective enrichment from occurring normally. Interpretation of neurological and behavioral dysfunction in mice in these circumstances can provide insight into how hippocampal circuits contribute to behavior and survival.

This work was supported by the Research Center Program of the Institute for Basic Science (IBS-R002-A1; M.W.J.).

P-37

Computational markers for successful prognosis in Nicotine Use Disorder: A longitudinal EMA study in real-world clinical settings

Jeung-Hyun Lee¹, Jaeyeong Yang¹, Sang Ho Lee¹, Hyeonjin Kim¹, Mark, A. Pitt², Jay, I. Myung², Hyung Jun Park³, Hee Kyoung Joh⁴, Woo-Young Ahn^{5*}

¹Department of Psychology, Seoul National University, ²Department of Psychology, The Ohio State University,

³Department of Family Medicine, Seoul National University Hospital, ⁴College of Medicine, Seoul National University,

⁵Department of Psychology, Department of Brain and Cognitive Sciences, Seoul National University)

Keywords: Addiction, Computational psychology, EMA

The efficacy of Nicotine Use Disorder (NUD) treatment remains limited, and the accurate prediction of successful cessation presents an ongoing challenge. Additionally, individual differences that prompt smoking decisions remain unclear. To address this gap, we conducted a longitudinal study using ecological momentary assessment (EMA) via smartphone application and decision-making tasks. Treatment-seeking smokers (N=87) engaged in a smoking cessation clinic that was assisted with medication for 5-6 weeks. Participants provided daily self-reports on psychological variables (i.e., anxiety, craving level for smoking, depression, mood, and stress) and completed two decision-making tasks powered by adaptive design optimization (ADO). The choice under risk and ambiguity (CRA) and delay discounting (DD) task, produced individual parameters of risk-taking, ambiguity-aversion, and impulsivity (i.e., discounting rate of delayed reward) based on computational models. A time-lagged prediction analysis demonstrated that the computational marker of ambiguity aversion was negatively predictive of daily consumption of cigarettes on the following day ($R^2=0.60$). In line with this, a logistic regression analysis revealed that individuals' ambiguity aversion assessed during the early phase of the clinic positively predicted prospective success in reduced amount of smoking ($\beta=1.17$, $OR=3.22$, AUC score: 0.750). The main effects of the two analyses survived after accounting for covariates such as medication intake and variance in daily subjective reports such as craving. The current results suggest that the application of computational modeling, on decision-making tasks offers insights to the identification of predictors of smoking behaviors. We expect that these findings could potentially pave the way for the development of individually-tailored interventions and enhance the prognostic capabilities of nicotine-cessation clinics.

We thank Yuzu Oh, Hoyoung Doh, Mina Kwon, Heesun Park, Yoseph Lee, Eunhwi Lee, Jooyeon Jamie Im, Suyeon Song, Ayeon Yim, Yoonseo Zoh, Soyeon Kim, Harhim Park, and Jihyun Hur for help with data collection, and all of the participants and staff at the Seoul National University Health Service Center.

P-38

Associated brain functional networks with major and minor hallucinations in Parkinson's disease

Yae Ji Kim¹, Kyoungwon Baik², Phil Hyu Lee^{2*}, Yong Jeong^{3*}

¹Program of Brain and Cognitive Engineering, Korea Advanced Institute of Science and Technology, ²Department of Neurology, Yonsei University College of Medicine, ³Department of Bio and Brain Engineering, Korea Advanced Institute of Science and Technology

Keywords: Parkinson's disease, hallucination, fMRI

Minor hallucinations (mHs) and well-structured major hallucinations (MHs) are frequently observed symptoms of Parkinson's disease (PD) psychosis. This study aimed to investigate the resting-state networks (RSNs) in three groups: PD patients without hallucinations (PD-nH), with minor hallucinations (PD-mH), and with major hallucinations (PD-MH).

A total of 73 PD patients were included in the study, with 27 in the PD-nH group, 23 in the PD-mH group, and 23 in the PD-MH group. By utilizing seed-based functional connectivity analyses, the researchers explored the RSNs that may be associated with hallucinations in PD, including the default mode network (DMN), executive control network (ECN), dorsal attention network (DAN), ventral attention network (VAN), and visual network (VN). Cognitive function and RSN connectivity were compared among the three groups. Additionally, a seed-to-seed analysis was performed to assess the inter-network connectivity within each group using the respective RSN seeds. PD-MH patients exhibited lower attention and visuospatial function test scores compared to the other groups. The connectivity of the right intracalcarine cortex within the DAN was lower in the PD-MH group than in the other groups. The PD-mH and PD-MH groups showed increased connectivity in the left orbitofrontal cortex within the DMN, while connectivity was decreased in the right middle frontal gyrus (MFG) within the ECN, precuneus cortex within the VAN, right middle temporal gyrus and precuneus cortex within the DAN, and left MFG within the VN. The PD-mH and PD-MH groups displayed distinct inter-network connectivity patterns among the five evaluated RSNs, particularly regarding DAN connectivity. Dysfunction in the DAN may play a crucial role in the progression from minor hallucinations to major hallucinations in PD patients.

This research was supported by the Basic Science Research Program of the National Research Foundation of Korea (NRF), funded by the Ministry of Science, ICT, and Future Planning (grant NRF-2019R1A2C2085462). Phil Hyu Lee receives funding from NRF-2019R1A2C2085462.

P-39

Getting personal: brain decoding of spontaneous thought using personal narratives

Hong Ji Kim¹, Byeol Kim Lux², Emily S. Finn², Choong-Wan Woo^{1*}

¹Department of Biomedical Engineering, Sungkyunkwan University, ²Department of Psychological and Brain Sciences, Dartmouth College

Keywords: Spontaneous thought, fMRI, predictive modeling

The contents and dynamics of spontaneous thought are important factors for personality traits and mental health. However, the assessment of spontaneous thought poses challenges due to their voluntary and unconstrained nature. Here, we aimed to decode two key content dimensions of spontaneous thought—self-relevance and valence—from brain activity. To train fMRI-based predictive models, we used individually generated personal stories as stimuli in the story-reading task to mimic narrative-like spontaneous thoughts ($n = 49$). The models were tested on two resting-state fMRI datasets ($n = 49$ and 90), achieving significant predictions. The default mode, ventral attention, and frontoparietal networks played key roles in the predictions, with the anterior insula and midcingulate cortex contributing to self-relevance prediction and the left temporoparietal junction and dorsomedial prefrontal cortex contributing to valence prediction. Overall, this study presents brain models of internal thoughts and emotions, highlighting the potential for the brain decoding of spontaneous thought.

This work was supported by IBS-R015-D1 (Institute for Basic Science; to C.-W.W.) and 2021M3E5D2A01022515 (National Research Foundation of Korea; C.-W.W.).

P-40

Stress responsiveness shapes the induction of synaptic plasticity in the lateral habenula

Hoyong Park, ChiHye Chung*

Department of Biological Sciences, Konkuk University

Keywords: Stress, Lateral habenula, Synaptic plasticity

Exposure to acute stress leads to abnormal activation of the lateral habenula (LHb), an epithalamic area important for stress responses. We previously observed dynamic changes in the synaptic efficacy of the LHb with great heterogeneity: a brief, strong theta burst stimulation (TBS) caused moderate long-term potentiation (LTP) only in a subpopulation of LHb neurons. Stress exposure significantly facilitated the induction of LTP in LHb synapses. Given that stress selectively activates certain LHb neurons, and the magnitude of potentiation varies among recorded neurons, we investigated whether stress-responsive neurons exhibit different responses to TBS compared to neurons that are not directly activated by stress. Interestingly, TBS in naïve mice led to a slight decrease in synaptic responses, likely through the activation of GABAB receptors. Using viral delivery of c-fos-tTA and TRE-EYFP, we successfully labeled stress-responsive neurons within the LHb and recorded excitatory postsynaptic currents (EPSCs) exclusively from these neurons. Upon TBS, the EPSC amplitudes of non-responsive neurons significantly decreased, while those of stress-responsive LHb neurons remained unchanged. Although the specific molecular or physiological characteristics that determine stress responsiveness are still unknown, our findings suggest that stress-responsive neurons are more readily potentiated or resistant to decreases in response to stimulation. This study, together with previous studies, unveils another mechanism contributing to the enhanced potentiation of the LHb following stress, offering valuable insights into understanding the hyperactivation of the LHb observed in animal models of depression.

NRF-2022R1C1C2008960

P-41

Effects of Multi-Echo EPI on Human Pain fMRI Experiment

Eui-Jin Jung^{1,2}, **Suhwan Gim^{1,2}**, **Choong-Wan Woo^{2,3*}**

¹Department of Global Bio-medical Engineering, Sungkyunkwan University, Suwon, Republic of Korea,

²Center for Neuroscience Imaging Research, Institute for Basic Science, Suwon, Republic of Korea,

³Sungkyunkwan University, Suwon, Republic of Korea

Keywords: Pain, Multi-Echo EPI, Predictive Model

The ventromedial prefrontal cortex (vmPFC) is a crucial region of interest in pain research, given its integral role in integrating sensory, social, and self-related information associated with pain processing (Roy et al., 2012). Notably, individual variability in neural representations of pain has been observed across multiple brain regions, including the vmPFC, wherein activation patterns diverge among individuals in response to identical painful stimuli (Kohoutová et al., 2022). However, functional magnetic resonance imaging (fMRI) has limitations such as susceptibility to noise and artifacts, particularly in frontal regions like the vmPFC. To address these challenges a study by Lynch et al. (2020) demonstrated that employing a multi-echo fMRI sequence can enhance the reliability of functional connectivity-based measurements during the resting state. We propose the use of a multi-echo fMRI sequence to enhance the detection of pain-related brain activations. Forty healthy participants (20 females) underwent 96 heat trials within the MRI scanner, comprising six temperature levels. Multi-echo fMRI scans were acquired utilizing a T2*-weighted echo-planar sequence, providing comprehensive coverage of the entire brain. The scanning parameters included a repetition time (TR) of 1,000ms, and echo times (TE) of 13.00ms, 31.26ms, and 49.52ms. For data preprocessing, we will employ TEDANA (TE Dependent ANALysis), a method specifically designed for denoising multi-echo fMRI data. Our primary analysis objective revolves around comparing the temporal signal-to-noise ratio (tSNR) across three distinct conditions: Single-echo, optimally combined, and ME-ICA. This assessment will involve the evaluation of tSNR across the entire brain, with a specific focus on pain predictive areas previously identified in the literature. Furthermore, our study aims to elucidate regions of interest exhibiting notable variance, encompassing the vmPFC and subcortical regions.

P-42

Sustained inhibitory long-term depression on the paraventricular nucleus of the hypothalamus in the absence of FKBP5

Seungjae Zhang, ChiHye Chung*

Department of Biological Sciences, Konkuk University

Keywords: BDNF-TrkB signaling, HPA axis, The Paraventricular nucleus of the hypothalamus (PVN)

The hypothalamic-pituitary-adrenal (HPA) axis governs stress-induced physiological responses. Stress exposure triggers the release of glucocorticoids, which then provide feedback signals to the brain to prevent excessive hormonal responses. The physiological model of this negative feedback is known as activity-dependent long-term depression in inhibitory synapses (iLTD) of the paraventricular nucleus (PVN) of the hypothalamus. FKBP5 is a well-known negative modulator of glucocorticoid receptors (GR), which plays a role in modulating stress-induced behaviors. Mice overexpressing FKBP5 have been reported to be susceptible to stress, exhibiting slow stress-coping behaviors and anxiogenic behaviors. On the other hand, FKBP5-deficient mice show resilience to stress and exhibit anxiolytic behaviors. Given that PVN-specific deletion of FKBP5 led to increased GR activity, we investigated the impact of FKBP5 deficiency on iLTD in the PVN. Interestingly, PVN neurons lacking FKBP5 exhibited iLTD even in the absence of GR activation, suggesting continuous negative feedback in the hypothalamus. The mechanisms underlying iLTD in the FKBP5-deficient PVN were found to involve the same signaling pathways as previously reported, including metabotropic glutamate receptor 5 and opioid receptors. Additionally, we discovered that the BDNF-TrkB signaling pathway also mediates iLTD in the PVN. Overall, these findings highlight the role of FKBP5 and the associated signaling pathways in regulating stress-induced behaviors and the negative feedback mechanism in the hypothalamus to maintain physiological balance during stress.

We thank all the Chung lab members for their help and discussions. The National Research Foundation of Korea supported this work, funded by the Ministry of Science, ICT, and Future Planning (NRF-2019M3C7A1031742, NRF-2020R1A2C2005868).

P-44

Validation of a modified sub-chronic variable stress animal model for studying sex differences in depression

Woonhee Kim, Sejong Oh, ChiHye Chung*

Department of Biological sciences, Konkuk University

Keywords: Depression, Sex difference, medial prefrontal cortex

Sex influences the development of mood disorders; however, current animal models mostly focus on male rodents, neglecting sex differences. Sub-chronic variable stress (SCVS) is a depression model that has been reported to induce depressive symptoms only in female mice but occasionally in males in our laboratory. In this study, we modified the SCVS model by reducing stress intensity to validate behavioral changes by using various tests related to depression. We found that the modified SCVS induced depressive symptoms in female C57BL/6 mice, while both male and female mice displayed anxiety-like behavior in the novelty-suppressed feeding test. Notably, further analysis of the tail suspension test (TST) using K-mean cluster analysis and receiver operating curve (ROC) analysis revealed increased mobility per bout, initial mobility, and overall immobility in females and brief but frequent struggling in males. We also examined the impact of the modified SCVS on the medial prefrontal cortex (mPFC) and found distinct changes depending on the subareas of mPFC. In the infralimbic cortex, SCVS-exposed males showed a higher firing threshold, impaired firing activity on strong stimuli, reduced excitatory amplitude, and increased inhibitory frequency. In the prelimbic cortex, only females exhibited increased excitability and reduced inhibitory frequency. Our study highlights the potential of the modified SCVS model to explore sex differences in depression, as evidenced by various criteria in the TST and alterations in mPFC activity. By considering both sexes, this model provides a valuable tool for further research on mood disorders.

This work was supported by the National Research Foundation of Korea (NRF-2019M3C7A1031742 and NRF-2020R1A2C2005868)

P-45

Spatial navigation training enhances hippocampal engagement and episodic memory

Jung Han Shin, Sang Ah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: spatial navigation, cognitive training, hippocampus

The hippocampus (HPC) is crucial for long-term memory formation and retrieval. Although the HPC is vulnerable to aging and neurodegenerative diseases, its structure and function change with experience, highlighting the necessity and potential of HPC-targeted cognitive training. Episodic memory (EM) is a complex HPC function that integrates what, where, and when, but it is challenging to train directly. Spatial navigation training has been studied more widely, but there is no evidence of its transferability to broader HPC functions like EM. In this study, we investigated this question using an EM task in which participants remembered objects (what) moving into boxes (where) in a specific order (when). 33 subjects (age: 24.1, 11 females) performed the task in the fMRI scanner before and after the training. The training consisted of 2 weeks of a spatial navigation task in which 17 subjects collected and retrieved a sequence of coins around a virtual space and then were provided with feedback on their performance. First, in the EM task, the spatiotemporal (where-when) binding was accompanied by higher engagement of HPC and was associated with better EM accuracy. Navigation training improved performance across all participants; however, spatial memory, partial EM components, and where-when binding was only enhanced in the feedback group. During EM, the HPC and cortical memory network activation increased with training, and a greater improvement in training led to greater engagement of the EM network, especially the subiculum. Training also increased functional connectivity between HPC and cortical regions, especially with the orbitofrontal cortex. Pattern reinstatement between encoding and retrieval was increased in both CA1 and subiculum. In summary, training basic navigational mechanisms enhanced EM accuracy and its neural correlates, extending existing knowledge of HPC plasticity and providing support for the development of navigation-based HPC memory training.

This work was supported by the National Research Foundation of Korea (2021M3E5D2A01023891 and 2023R1A2C2005587), Creative-Pioneering Researchers Program of Seoul National University, and Samsung Electronics (A0426-20220108)

P-46

Heterogeneity in Stress Response to COVID-19 Related Discrimination in Children: Insights from the ABCD Study

Jinwoo Yi¹, Eunji Lee², Bogyom Kim², Gakyung Kim¹, Yoonjung Yoonie Joo³, Jiook Cha^{2*}

¹Department of Brain and Cognitive Sciences, Seoul National University, ²Department of Psychology, Seoul National University, ³Institute of Data Science, Korea University

Keywords: heterogeneity, COVID-19-related discrimination, frontotemporal structure

Experiences of COVID-19-related discrimination in children may lead to acute and long-term effects on their stress levels in an individualized manner. Considering that adversity during the neurodevelopmental stage may have long term sequelae of psychopathology, it is crucial to identify the factors rendering certain children particularly more vulnerable or resilient to stressful events. However, prior findings are mixed on the risk and protective factors including children's socioeconomic, neurobiological, and genetic characteristics. In this study, we investigated the acute and long-term effects of COVID-19-related discrimination on the stress levels and their risk and protective factors in 1,102 preadolescents through the Adolescent Brain and Cognitive Development (ABCD) study. Using Generalized Random Forest (Athey & Wager, 2019), allowing for robust estimation of causal effects of the treatment and evaluation of heterogeneity via quasi-experimental simulation, we tested the individual differences of stress responses to COVID-19 related discrimination and sought for major factors highly associated with this heterogeneity among socioeconomic, behavioral, neurobiological and genetic features. Experiences of discrimination had significant acute and long-term effects on preadolescents' stress levels. While the acute effects were relatively homogeneous across children, the long-term effects were more heterogeneous. We identified the frontotemporal structural connectivity and cortical volume as particularly important factors more closely correlated with individual differences in long-term effects with long-term effects compared to other features. This study highlights the detrimental impact of discrimination during the pandemic and the importance of the frontotemporal structures in preadolescent stress sensitivity. By focusing on heterogeneity and its key factors, our findings have the potential for prescriptive modeling to prevent children's future psychopathology.

This work was supported by the New Faculty Startup Fund, Creative Pioneering Researchers Program through Seoul National University, The National Research Foundation of Korea grants, and the BK21 FOUR programs funded through the National Research Foundation of Korea.

P-47

Development of fronto-hippocampal network function in episodic memory during adolescence

Kahyun Choi¹, Jiook Cha², Dongil Chung³, Grace Chung⁴, Jae Won Kim⁵, Sang Ah Lee^{1*}

¹Department of Brain and Cognitive Sciences, Seoul National University, ²Department of Psychology, Seoul National University; Department of Brain and Cognitive Sciences, Seoul National University; AI Institute, Seoul National University, ³Department of Biomedical Engineering, UNIST, ⁴Department of Child Development & Family Studies, Seoul National University, ⁵Division of Child and Adolescent Psychiatry, Department of Psychiatry, Seoul National University College of Medicine; Department of Psychiatry, Seoul National University Hospital

Keywords: Episodic memory, Adolescent development, Fronto-hippocampal connectivity

The hippocampus (HPC) and the prefrontal cortex (PFC) undergo developmental changes across childhood and adolescence. However, studies have reported mixed results on changes in the subregions of the HPC and its functional connectivity with the PFC, and their effects on memory performance. In the present study, we investigated hippocampal memory in a sample of eighty adolescents, ranging from 10 to 17 years of age. Neural activity was measured using fMRI while participants performed an episodic memory task that required them to remember what animals (what) appeared at which locations (where) in what order (when) on each trial.

We found both age-related and memory-related correlates of episodic memory. During the encoding session, activation in posterior HPC was greater than in anterior HPC, and this anterior-posterior difference increased with age, driven by a decrease in activation in the anterior HPC. Distinct developmental changes in connectivity patterns with different cortical regions were observed with the anterior and posterior HPC as the seed region. Specifically, connectivity between the anterior HPC and frontal regions was correlated with episodic memory performance in both younger (age<15) and older (age>15) adolescents but showed distinct patterns. Moreover, greater activation in the right HPC compared to the left HPC, particularly in the hippocampus tail, was associated with better performance in general. These findings suggest that functional specialization of the hippocampus along the anterior-posterior axis, and the parallel development of the fronto-hippocampal network, may underlie the development of episodic memory during adolescence.

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2021M3A9E4080780, and No. 2023R1A2C2005587), Samsung Electronics Co., Ltd (A0426-20220108), and SNU Center for Happiness Studies (No. 0404-20220001).

P-48

SwiFT: Swin 4D fMRI Transformer

Junbeom Kwon¹, Yongho Kim², Sangyoon Bae³, Sunghwan Joo⁴, Donggyu Lee⁴, Yoonho Jung², Shinjae Yoo^{5*}, Jiok Cha^{1*}, Taesup Moon^{2*}

¹Department of Psychology, Seoul National University, ²Department of Electrical and Computer Engineering, Seoul National University, ³Interdisciplinary Program in Artificial Intelligence, Seoul National University, ⁴Department of Electrical and Computer Engineering, SungKyunKwan University, ⁵Brookhaven National Lab

Keywords: Spatiotemporal brain dynamics, 4D fMRI Transformer, Prediction of biological and cognitive variables

The modeling of spatiotemporal brain dynamics from high-dimensional data, such as 4D functional MRI, is a formidable task in neuroscience. To address this challenge, we present SwiFT (Swin 4D fMRI Transformer), a Swin Transformer architecture that can learn brain dynamics directly from 4D functional brain MRI data in a memory and computation-efficient manner. SwiFT achieves this by implementing a 4D window multi-head self-attention mechanism and absolute positional embeddings. We evaluate SwiFT using multiple largest-scale human functional brain imaging datasets in tasks such as predicting sex, age, and cognitive intelligence. Our experimental outcomes reveal that SwiFT consistently outperforms recent state-of-the-art models. To the best of our knowledge, SwiFT is the first Swin Transformer architecture that can process dimensional spatiotemporal brain functional data in an end-to-end fashion. Furthermore, due to the end-to-end learning capability, we also show that contrastive loss-based self-supervised pre-training of SwiFT is also feasible for achieving improved performance on a downstream task. We believe that our work holds substantial potential in facilitating scalable learning of functional brain imaging in neuroscience research by reducing the hurdles associated with applying Transformer models to high-dimensional fMRI.

I would like to express my sincere gratitude and appreciation to NESAP (Exascale Science Applications Program) teams, who provided me with a conducive research environment and access to valuable GPU resources, Perlmutter in NERSC. The paper is currently undergoing the review process for the NeurIPS 2023 conference

P-49

Age-related Impairments in Landmark Utilization during Spatial Memory Retrieval

Jeonghyun Lee, Sang-Eon Park, Sang Ah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: Spatial Navigation, Aging, EEG spectral similarity

Past studies reported that, in contrast to a relative preservation of egocentric spatial memory, there is a decline in landmark processing in aging. To investigate the electrophysiological correlates underlying age-related impairments in landmark-use, we conducted a task in which participants watched 1-min videos of navigation in virtual environments and then subsequently identified overhead maps of the traveled path. We measured scalp EEG and eye-gaze during navigation and map-viewing (retrieval) in 27 younger (age 20-45) and 21 older adults (age 45-65).

If age-related decline in spatial memory is driven by difficulties in retrieving landmark information, we hypothesized that the EEG correlates for landmark processing during navigation will reactivate during map-viewing in young adults but to a lesser degree in older adults. First, we found that posterior regions showed higher theta (3-6Hz) and lower beta (15-21Hz) power during landmark gazing during the navigation encoding phase. We correlated EEG spectral profiles between timepoints in which participants were gazing at landmarks during navigation with the EEG spectral profile during the map-viewing phase. In younger participants, we found higher spectral similarity for timepoints related to landmark-viewing than other timepoints, especially in the posterior channels. We also found higher similarity for the spectral profile during turning (at a corner) than going straight. In older participants, however, we only found higher similarity for turning but not for landmark-viewing. Spectral similarity was generally higher for correct trials than incorrect trials. These results suggest that similarity in neural activation patterns between spatial memory encoding and retrieval was modulated both by the specific navigational event (e.g., landmark-viewing, turning) and memory retrieval success and that older adults show specific impairments in reactivating landmark-related neurocognitive processes to guide their memory.

This study was supported by the Creative-Pioneering Researchers Program through Seoul National University and the National Research Foundation of Korea (2021M3E5D2A01023891 and 2020K1A3A1A19088932).

P-50

Exploring the Role of White Matter Microstructure as a Mediator in the Relationship Between PTSD Severity and Social factors in Adolescent Girls

Seoyoung Lim¹, Ana Carolina Coelho Milani², Jiook Cha^{3*}

¹Department of Brain and Cognitive Sciences, Seoul National University, ²Departamento de Psiquiatria, Universidade Federal de Sao Paulo, R Rua Major Maragiliano, SP, Brazil, ³Department of Psychology, College of Social Sciences, Seoul National University

Keywords: dMRI, Mediation, PTSD

Insight into the neural changes and the influence of environmental risk factors over time in adolescents with Post-traumatic stress disorder (PTSD) can elucidate the onset of the disorder and inform effective therapeutic strategy development. Here we aimed to explore how early-life social factors and maltreatment might intensify PTSD symptoms, and how these factors contribute to whole brain fractional anisotropy (FA) and FA changes in 18 adolescent girls with PTSD due to sexual assault (mean age : 15.7 ± 1.04 years) and 21 healthy controls (all adolescent girls ; mean age : 16.2 ± 1.21 years) in São Paulo, Brazil, using diffusion tensor imaging (DTI) , and PTSD symptom measures at baseline. Through the mediation analysis, we discovered that FA in the internal capsule and uncinate fasciculus mediated the impact of emotional abuse and parental maltreatment on PTSD symptoms. Specifically, the internal capsule played a pivotal role, with emotional abuse showing an Average Causal Mediated Effect (ACME) of 0.014 (95% CI, [-1.776 , -0.16]; corrected p-value = 0.014), while parental maltreatment exhibited an ACME of 0.042 (95% CI, [-0.2264 , -0.4806]; corrected p-value = 0.042). Additionally, the anterior limb of the internal capsule showed significant differences between pre- and post-treatment groups in a Tract-Based Spatial Statistics (TBSS) analysis. In the TBSS analysis, when contrasting the Healthy Control (HC) group with the Pre-treatment PTSD group, and further comparing the same subjects from the Pre-treatment to Post-treatment phases within the PTSD group, there were significant t-values of 2.5724 and 6.305 (corrected p-value < 0.05), respectively. Considering that the Internal Capsule is associated with sensory fibers integral to perception and the regulation of motor functions, it is plausible that emotional abuse and parental maltreatment may have an influence on the neural pathways responsible for processing stimuli and responses associated with trauma.

P-51

The effect of gamified emotional attention-training tasks on cognitive bias, affective health, and resting-state EEG markers

Jisu Chung, Minsuk Koo, Minwoo Kim, Sang Ah Lee*

Department of Brain and Cognitive Sciences, Seoul National University

Keywords: neurocognitive training, emotional attentive bias, digital health

Emotional attention bias refers to the tendency of individuals to selectively attend to and process emotionally salient information. It plays a crucial role in cognitive function and emotional well-being. Based on previous research that showed that attentional biases can change with training, the present study aimed to investigate the effects of various emotional-attentional training tasks on cognitive bias, affective health. Furthermore, in order to understand the neural correlates of those changes, we also recorded prefrontal EEG during resting-state, before and after the training. The pre- and post-tests included three attentional tasks: an emotional flanker task, an emotional feature-conjunction task, and a social approach and avoidance task; the stimuli consisted of graphic animal faces. The participants (training group $n = 22$; 10 female, control group $n = 19$; 8 female) completed a total of four training sessions using gamified versions of the attentional task over a period of two weeks, with each session lasting for 30 minutes.

The results showed that repeated engagement in attentional tasks with emotional stimuli led to improvements in overall reaction time across all three tasks. While not all participants showed improvements in psychological well-being, there were significant correlations observed between the degree of improvement in depressive and anxiety symptoms and cognitive bias in the task. Additionally, training was associated with improvements in prefrontal EEG asymmetry and showed a significant relationship with task performance. These findings have implications for understanding the role of attentional bias in emotional health and highlight the potential benefits of digital attentional training in improving cognition and emotional well-being.

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2023R1A2C2005587 and No. 2021M3A9E4080780) and Samsung Electronics Co., Ltd (A0426-20220108).

P-52

Novelty exposure enhances immediate extinction memory and the inhibition of the BLA does not hinder the effects of novelty exposure

Bomi Kim, June-Seek Choi*

Department of Psychology, Korea University

Keywords: immediate extinction, novelty exposure, basolateral amygdala

Exposure to a novel event has been recently studied for one of the interventions that facilitate consolidation of fear extinction memory. This behavioral procedure is based on the principle that hippocampal synapses related to the previous extinction learning get strengthened, during their processing of information about a novel event. However, it is not yet known whether such interventions also produce the same effects in immediate extinction conditions, where fear conditioning takes place and extinction learning occurs shortly thereafter. Here, we investigated whether the exposure to novel open field could induce the successful retrieval of extinction memory in the immediate extinction paradigm. The group of Sprague-Dawley rats that explored a novel open field for 5 minutes after contextual fear conditioning (n=6, Novelty Group) demonstrated the persistence of long-term memory for immediate extinction in a test conducted 48 hours after conditioning. On the other hand, the group staying in the home cage for the same time (n=6, Home cage Group) significantly showed more fear responses than Novelty Group in the test. Additionally, to investigate the contribution of the BLA to these effects which is known for conveying spatial and contextual information to the hippocampus, BLA was optogenetically inhibited during open field exploration. The group of which BLA was inhibited during open field exploration (n=8, NE+BI Group) exhibited better maintenance of immediate extinction memory compared to the group in which BLA was not inhibited (n=8, NE+BDI Group), suggesting that inhibiting the activation of the BLA during a novel event does not interrupt the novelty-based memory enhancement. These results demonstrate that exposure to a novel event enhances extinction memory even in immediate extinction conditions and could be a promising behavioral intervention for the treatment of traumatic fear memories, while the BLA does not significantly contribute to these effects.

This work was supported by the National Research Foundation of Korea (NRF) grant and the Brain Convergence Research Program (2021M3E5D2A01023887) funded by the Korean Government (MSIT).

P-53

Cortico-hippocampal correlates of aging and age-related spatial memory decline using intracranial EEG

Sang-Eon Park¹, Thomas Donoghue², Joshua Jacobs², Sang Ah Lee^{1*}

¹Department of Brain and Cognitive Sciences, Seoul National University, ²Department of Biomedical Engineering, Columbia University

Keywords: iEEG, spatial memory, aging

Previous studies have associated age-related functional degeneration of the hippocampus with spatial memory impairment, but the underlying neurophysiological mechanisms have been elusive due to the co-occurrence of general age-related changes. Intracranial EEG (iEEG) recordings provide a deeper understanding of the human brain by identifying direct electrophysiological correlates of cognitive function. We used iEEG recordings across the whole brain to investigate age-related neural changes in 69 presurgical epilepsy patients (19 to 61 years of age) while they performed a computer-based spatial navigation task. In order to analyze periodic and aperiodic components separately, we first isolated periodic narrowband power by removing the aperiodic broadband component (1/f spectral slope) from the iEEG signal by applying the FOOOF (fitting oscillations & 1/f) algorithm. In the aperiodic component, we found a flatter 1/f slope (associated with an excitatory state) in the prefrontal cortex with age and, in contrast, a steeper 1/f slope (indicating an inhibitory state) in the hippocampus. In the periodic component, older adults (age>44) showed decreases in theta (3-6Hz) power in the hippocampus. Older adults also showed increased alpha (6-10Hz) and beta (12-18Hz) and decreased gamma (37-50Hz) power in various brain regions such as the hippocampus, frontoparietal and temporal cortex but not in the prefrontal cortex. When examining the neural markers related to spatial memory accuracy, we found that the decrease in hippocampal theta was more pronounced in low-performing older adults. On the contrary, high-performing older adults had a flatter aperiodic 1/f slope, specifically in the DLPFC. These results show both an age-related degeneration of hippocampal memory function, as indicated by the weakened theta oscillation in low-performing subjects, alongside a potential compensatory mechanism in the older adults through the activation, indicated by changes in 1/f, of the DLPFC.

This study was supported by Creative-Pioneering Researchers Program through Seoul National University and the National Research Foundation of Korea (2021M3E5D2A01023891 and 2020K1A3A1A19088932).

P-54

Transcranial Focused Ultrasound stimulation in the infralimbic cortex enhances fear extinction in rats

Jae Yong Lee¹, Ye Eun Kim¹, Jihong Lim², Hyunjoo Jenny Lee^{2*}, Yong Sang Jo^{1*}, June-Seek Choi^{1*}

¹School Of Psychology, Korea University, ²School Of Electrical Engineering, KAIST

Keywords: Transcranial Focused Ultrasound, Infralimbic Cortex, Fear Extinction

Fear extinction, a reduction in fear response to a threat-associated cue when repetitively presented without threat, is a crucial adaptive process and its failure could result in various fear-related disorders. Throughout numerous brain regions, the infralimbic cortex (IL) has been extensively studied for its role in modulating fear extinction. With the latest non-invasive neuromodulatory technique, Transcranial Focused Ultrasound (tFUS), which has a high spatial resolution, our study investigated whether the tFUS stimulation in the IL enhances fear extinction. Custom-made tFUS or dummy headgears targeting the IL were surgically implanted above the skull in twenty-two male Sprague-Dawley rats, one week prior to the experiment. On day 1, tone stimulus (80dB, 4KHz, 30 s) was presented 5 times in advance to eliminate innate responses to the tone. The conditioning procedure was followed immediately and both tFUS (n=11) and control (n=11) groups were given 5 pairings of tone and coterminating footshock (0.7mA, 0.5 s). On day 2, rats underwent extinction trials, given 8 tones without footshock, while tFUS stimulations (5MHz, 30 s) were given simultaneously during tone presentation in the tFUS group. Extinction memory was tested after 24 hours (Day 3) by presenting the tone 2 times. The tFUS group exhibited significantly lower freezing levels compared to the control group in extinction sessions and lower freezing levels remained in the retention test, indicating facilitated fear extinction memory. Additional open field tests ruled out locomotive changes of the tFUS stimulation. These results collectively suggest that the tFUS stimulation of the IL enhances fear extinction.

This work was supported by the National Research Foundation of Korea grant (2021M3E5D2A01023887 to J.-S.C. and 2022M3E5E8017804 to Y.S.J.) funded by the Korean Government (MSIT).

P-55

Development of memory binding in preschool-aged children: A focus on object-space and time-space associations

Gakyung Kim¹, Yujin Rah¹, Hyewon Jun², Sang Ah Lee^{1*}

¹Department of Brain and Cognitive Sciences, Seoul National University), ²Department of Brain and Cognitive Sciences, Ewha Womans University

Keywords: episodic memory, cognitive development, object-space association

In daily life, the comprehensive binding of what happened, when, and where is crucial for accurate event memory. The maturation of the hippocampal formation plays a pivotal role in the development of these memory processes. To investigate the progression of memory binding in early childhood, we conducted a study with 81 Korean preschool-aged children (32 girls, aged 36-83 months) in which we devised a nonverbal object placement task for the distinct assessment of 'what' (different objects), 'when' (order of placement), and 'where' (locations of cups) information. In each trial, the child watched as the experimenter hid three distinct objects into three of the five cups on a tabletop apparatus; children were then asked to re-enact the sequence after performing an interference task. We contrasted the binding of 'where' information with 'what' and 'when', specifically, by giving the objects to the children during the re-enactment phase either in the correct order or in the wrong order. Overall, children showed spatial memory improvements with age. However, younger children also tended to make more errors according to the sequence of locations in the latter condition, showing a weaker binding of the object with its location. By 5 years of age, children placed the objects into the correct cups with high accuracy even when provided in the wrong order. The stronger influence of when-where binding in younger children suggests that the memory of spatial sequences develops early; but around 5 years of age, a stronger bias towards object-space associations emerges, potentially playing an important role in the maturation of adult-like episodic memory.

We would like to thank Hyeji Kim, Jina Lyu, and Inhee Kwak for their assistance in running the experiments. This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP; No. 2019-0-01371-003), the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT; No. 2023R1A2C2005587), and SYEDU (Multisensory Learning and Memory).

P-56

Dominance and Risk Regulate Prosocial Behavior in Rats

Subin Yang, In-Hyun Baek, June-Seek Choi*

Department of Psychology, Korea University

Keywords: Prosocial, Risk, Dominance, Restrainer

Although dominance and cost of helping behavior are two of the well-known factors for prosocial behavior, the interplay between these factors and their impact on behavior remains unexplained. In this research, the extent to which dominance regulates prosocial behavior according to the cost of helping behavior was measured. First, we employed the dominant tube test to evaluate the degree of dominance between cagemate rats. We determined dominance by analyzing the behavior of two rats as they simultaneously started from each end of a circular tube (6 x 60 cm) and pushed each other out. To estimate prosocial behavior, we divided the situations in which one of the rats (helper) helps the other rat in a restrainer (11 x 31 x 14 cm) into three conditions: no-cost (NC, opening the restrainer with a simple approach, n = 24), low-cost (LC, opening the restrainer with single lever pressing, n = 48), and high-cost (LC + the presence of a moving robot that imitates risk, n = 24). Initially, the helpers were trained to open the door for 2 days (NC, LC) or 4 days (HC) in the absence of a restrained rat, following which the rescue behavior was measured over the next 5 days in the presence of a restrained rat. We observed that a significant effect of dominance difference was only found in the HC condition. In particular, the latency of helping behavior was found to be significantly shorter in the dominant group compared to the subordinate group, despite the presence of the risk-imposing robot. However, no significant difference between the dominant and subordinate groups was observed in either the NC or LC conditions. Our results therefore demonstrate that dominance facilitates the prosocial behavior only in the presence of perceived risk. This kind of dominance effect in a limited context on prosocial behavior offered insights into the intricate interplay between risk and dominance in shaping prosocial behavior.

This study was supported by the National Research Foundation of Korea (NRF) grant (20212M3E5D2A01023887).

P-57

Autism and social deficit: Investigating how ‘self’ and ‘others’ are encoded in the autistic rodent brain

Eunkyu Hwang¹, Woohyun Kim², Soogeun Shin¹, Sanghee Lee¹, Eunjoon Kim², Eunee Lee^{1*}

¹Department of Anatomy, Yonsei University Medical School, ²Center for Synaptic Brain Dysfunctions/Korea Advanced Institute of Science and Technology

Keywords: Autism, Social interaction, In-vivo electrophysiology

Social interaction is a coordinated activity between oneself and others that involves processing of various types of sensory information originating from the self and the others. Patients diagnosed with autistic spectrum disorder (ASD) are often incapable of engaging in a typical social interaction and show a lack of social interest or motivation. These individuals are often described to be unable to appropriately interpret the state and experience of others. They may also have a heightened sensitivity and preoccupation with self. The unique characteristics of the neural activities in ASD patients that may lead to this specific quality of social deficit remain elusive. Recent literature highlights the importance of self and other discriminability in normative social behavior. This refers to the ability to differentiate the agency of a specific social experience, such as approaching, between the self and other. We hypothesize that individuals with ASD are unable to distinguish to whom such experience belongs, resulting in an atypical neural representation of self and other, thus causing social deficits. To address this hypothesis about the disrupted neural encoding of self and other, we conducted simultaneous in vivo electrophysiology recordings in the medial prefrontal cortex (mPFC) of socially interacting mice, a brain region that has been found to play an essential role in social behaviors. We employ two types of mice, Shank2 gene knockout mice (Shank2 KO), a well-established animal model of ASD and Shank2 wildtype mice (Shank2 WT). We employ two types of behavioral assays, the direct interaction test and the tube test. We present behavioral findings, and results from single neuron neuronal correlates analysis, and local field potential analysis. Preliminary results suggest genotype-specific neural activities in the mPFC of socially interacting mice.

Funding: NRF-2019R1A2C1084812 / HI22C1234

P-58

Effects of the chemogenetic lateral habenula activation on the association of CS with non-reward in Pavlovian appetitive conditioning

In-Beom Jin, Sang-Pil Park, Jung-Soo Han*

Department of Biological Sciences, Konkuk University

Keywords: lateral habenula , conditioned inhibition, chemogenetics

The presentation of an unconditioned stimulus (US) contingent upon the occurrence of a conditioned stimulus (CS) establishes excitatory conditioning or conditioned excitation, such as approaching a food-cup area. Conversely, associating a CS with the nonoccurrence of a US generates inhibitory conditioning or conditioned inhibition, such as not approaching a food-cup area. While previous research has predominantly focused on conditioned excitation, less attention has been given to the neural mechanisms underlying conditioned inhibition. In a previous study, our objective was to demonstrate the effects of chemogenetic inactivation of the lateral habenula (LHb) inactivation on the establishment of conditioned inhibition. Inhibiting LHb activity during unpaired training eliminated the delayed acquisition of subsequent excitatory learning following inhibitory learning. However, LHb inhibition during paired training had no effect. In the present study, we examined whether chemogenetic LHb activation facilitates the association of CS with the nonoccurrence of US. We adapted the same experimental design described above and employed the chemogenetic method to test it. In this study, rats went through 3 days of unpaired training, which did not retard the acquisition of the subsequent excitatory conditioning, long with chemogenetic LHb activation. And then paired training was followed. Rats with chemogenetic LHb activation and the 3-day unpaired training exhibited delayed acquisition of subsequent excitatory conditioning. These results suggest that LHb activation promotes inhibitory learning. These findings contribute to our understanding of the neural mechanisms underlying conditioned inhibition and shed light on the role of the LHb in this process.

Supported by the Korea Research Foundation Grant funded by the Korean Government (2020M3E5D9080734 to Jung-Soo Han).

P-59

Cell type-specific SCN2A haploinsufficiency induces epileptic seizures and social deficits

Eun-Hwa Hong, Haeun Noh, Eunkyu Hwang, Jaewon Han, Eunee Lee*

Department of Anatomy, Yonsei University College of Medicine

Keywords: epilepsy, autism, SCN2A

Mutations in the SCN2A gene, which encodes the neuronal voltage-gated sodium channel Nav1.2, are associated with epilepsy, intellectual disability, and autism (Ogiwara et al., 2018; Shin et al., 2019). Especially, loss-of-function mutations in the SCN2A gene cause autism with later-onset epileptic seizures, whereas gain-of-function mutations cause early-onset epileptic seizures. Despite the amount of research on SCN2A mutations, it is still not clear whether the abnormal behavioral phenotypes of SCN2A haploinsufficiency are more strongly correlated with excitatory or inhibitory neurons.

In this study, to investigate the effect of SCN2A deletion in excitatory or inhibitory neurons on SCN2A haploinsufficiency, SCN2A conditional knockout mice were generated using Emx1-Cre recombinase-mediated (Emx1-Cre) excitatory neuron-specific deletion or Viaat-Cre-recombinase-mediated (Viaat-Cre) inhibitory neuron-specific deletion.

Additionally, the series of behavioral and electrophysiological experiment were conducted to investigate the characteristics of the SCN2A loss-of-function mutations. Preliminary results suggest that SCN2A haploinsufficient mice showed a significant increase in social interaction behavior compared to wild type. Interestingly, SCN2A^{flox/+}/Emx1-Cre mice and SCN2A^{flox/+}/Viaat-Cre mice exhibited different behavioral phenotypes during the social interaction test. Moreover, the mice exhibited seizure-like activity during the electroencephalography (EEG) recordings.

This research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare, Republic of Korea (grant number: HI21C1659)

P-60

Chemogenetic inactivation of the lateral habenula impairs the association of a CS with the absence of an US, as assessed by the superconditioning task

Nam-Heon Kim, Su-Yeon Sim, Jung-Soo Han*

Department of biological science, Konkuk University

Keywords: Lateral Habenula, Pavlovian appetitive conditioning, Conditioned inhibition

The present study aimed to explore the neural substrates involved in conditioning inhibition, which occurs when a conditioned stimulus (CS) is associated with the absence of an unconditioned stimulus (US). Recent research suggests that the lateral habenula (LHb) plays a role in the association of a CS with the nonoccurrence of a US. To investigate this further, we conducted experiments using chemogenetic inhibition of the LHb to examine its effect on the inhibitory property of a CS generated by explicitly unpaired CS and US procedures. We employed the superconditioning task to assess the inhibitory property of CS, which consisted of three phases. During the initial phase, animals underwent six days of explicitly unpaired training of a light (CS1) with food in the behavioral chamber. In the subsequent phase, the animals received six days of paired training involving a light (CS1)-tone (CS2) compound paired with food. In the final phase, food-cup responses to each CS were measured over two days, with four trials of each CS per day. A comparison group of rats only underwent the second phase of the paired training, followed by the third phase. The results revealed a significantly higher food-cup response to the tone (CS2) was significantly higher in animals that underwent explicitly unpaired training during the first phase, compared to the comparison group. These findings demonstrate the successful generation of conditioned inhibition through explicitly unpaired training. Additionally, we investigated the effects of experimental manipulations throughout the explicitly unpaired training on the conditioned inhibitory properties. Animals with chemogenetic LHb inhibition throughout the unpaired training showed diminished conditioned inhibitory properties. These results indicate that chemogenetic LHb inhibition impairs the association of a CS with the absence of a US, thereby supporting the notion that LHb mediates a conditioned inhibition.

Supported by the Korea Research Foundation Grant funded by the Korean Government (2020M3E5D9080734 to Jung-Soo Han).

Electrophysiology Multi-Channel In-vivo Recording System



- > **DigitalLynx**
최대 512channel, 40kHz로 동시에 sampling 가능한 고성능 data acquisition system.

High-density, DC-coupled acquisition with closed loop processing



- > **Freelynx**
무선 headstage로부터 전송되는 시그널을 최대 10미터 거리에서 256channel까지 무선으로 데이터 취득 가능한 system.

Reliable, configurable, wide bandwidth wireless acquisition



- > **Lablynx**
head stage와 유선으로 연결된 소형화된 data acquisition system으로 취득한 데이터를 무선으로 PC로 전송하는 이동형 system. 최대 256channel 사용가능.

Low cost, portable, battery-powered wide bandwidth digital acquisition

med associates inc 행동실험 분석장비

> Operant Conditioning and General Behavior

> Self Administration System

> Shuttle Box



(주) 이 우 과 학 교 역
IWOO Scientific Corporation

서울시 서초구 반포대로 23길 30, 5층 www.iwoo.co.kr
Tel. 02-3473-2332 E-mail. sale@iwoo.co.kr

Amyloid Peptide

루젠SCI 공식 대리점 바이오토리 추가할인 이벤트

Cosmetics or Generic peptide 대량 생산 기반 구축
논문에 검증된 최고의 품질 최저가 보장

Custom Peptide Bulk 생산 가능

biostory@bio-story.com 031-283-1800

BIO STORY

(주)바이오 스토리



Lugen Sci Co., Ltd.

Product Name	β -Amyloid (Human, 1-42)
Catalog	AB142
Molecular Weight	4514.07
Sequence	DAEFRHDSGYEVHHQKLVFF AEDVGSNKGAIIGLMVGGVVI A

Product Name	β -Amyloid (Human, 1-40)
Catalog	AB140
Molecular Weight	4329.9
Sequence	DAEFRHDSGYEVHHQKLVFF AEDVGSNKGAIIGLMVGGVV

Product Name	β -Amyloid (Human, 25-35)
Catalog	AB2535
Molecular Weight	1060.4
Sequence	GSNKGAIIGLM

Product Name	β -Amyloid (Human, 42-1)
Catalog	AB421
Molecular Weight	4514.07
Sequence	AIVVGGVMLGIAGKNSGVDE AFFVLKQHHVEYGSDHRFEA D

Product Name	β -Amyloid (Human, 1-28)
Catalog	AB128
Molecular Weight	4329.9
Sequence	DAEFRHDSGYEVHHQKLVFF AEDVGSNK

Product Name	β -Amyloid (Human, 25-35), scrambled
Catalog	AB2535s
Molecular Weight	1060.4
Sequence	MAKGINGISGL

Citations

Epitope Mapping Immunoassay Analysis of the Interaction
between β -Amyloid and Fibrinogen Int J Mol Sci.
2019 Feb; 20(3): 496. doi: 10.3390/ijms20030496

Urolithin A suppresses high glucose-induced neuronal
amyloidogenesis by modulating TGM2-dependent
ER-mitochondria contacts and calcium homeostasis. Cell Death
Differ (2021) 28:184-202. doi: 10.1038/s41418-020-0593-1

Neurosplash

2023 한국뇌신경과학회 고등인지분과 하계 워크숍



한국뇌신경과학회
The Korean Society for
Brain and Neural Sciences

ibS 기초과학연구원
Institute for Basic Science



Brain & Cognitive Sciences
서울대학교 뇌인지과학과

인쇄일 2023년 8월 14일
발행일 2023년 8월 20일

발행처 한국뇌신경과학회, 기초과학연구원
[04378] 서울시 용산구 한강대로 95, 래미안용산 더 센트럴 A동 721호
Tel. 02-871-1862,1863, Fax. 02-790-1862, E-mail. neuro@ksbns.org

편집제작 (주)더 위드인
[07270] 서울시 영등포구 양산로 43 (양평동3가 16) 양평동우림이비지센타2차 1005호
Tel. 02-6959-5333, Fax. 070-8677-6333, E-mail. with@thewithin.co.kr

Neurosplash

**2023 한국뇌신경과학회
고등인지분과 하계 워크숍**



한국뇌신경과학회
The Korean Society for
Brain and Neural Sciences



기초과학연구원
Institute for Basic Science



Brain & Cognitive Sciences
서울대학교 뇌인지과학과